

A FRIENDLY INTRODUCTION TO COSMOLOGY

RENAN ISQUIERDO BOSCHETTI

WHY IS COSMOLOGY INTERESTING?

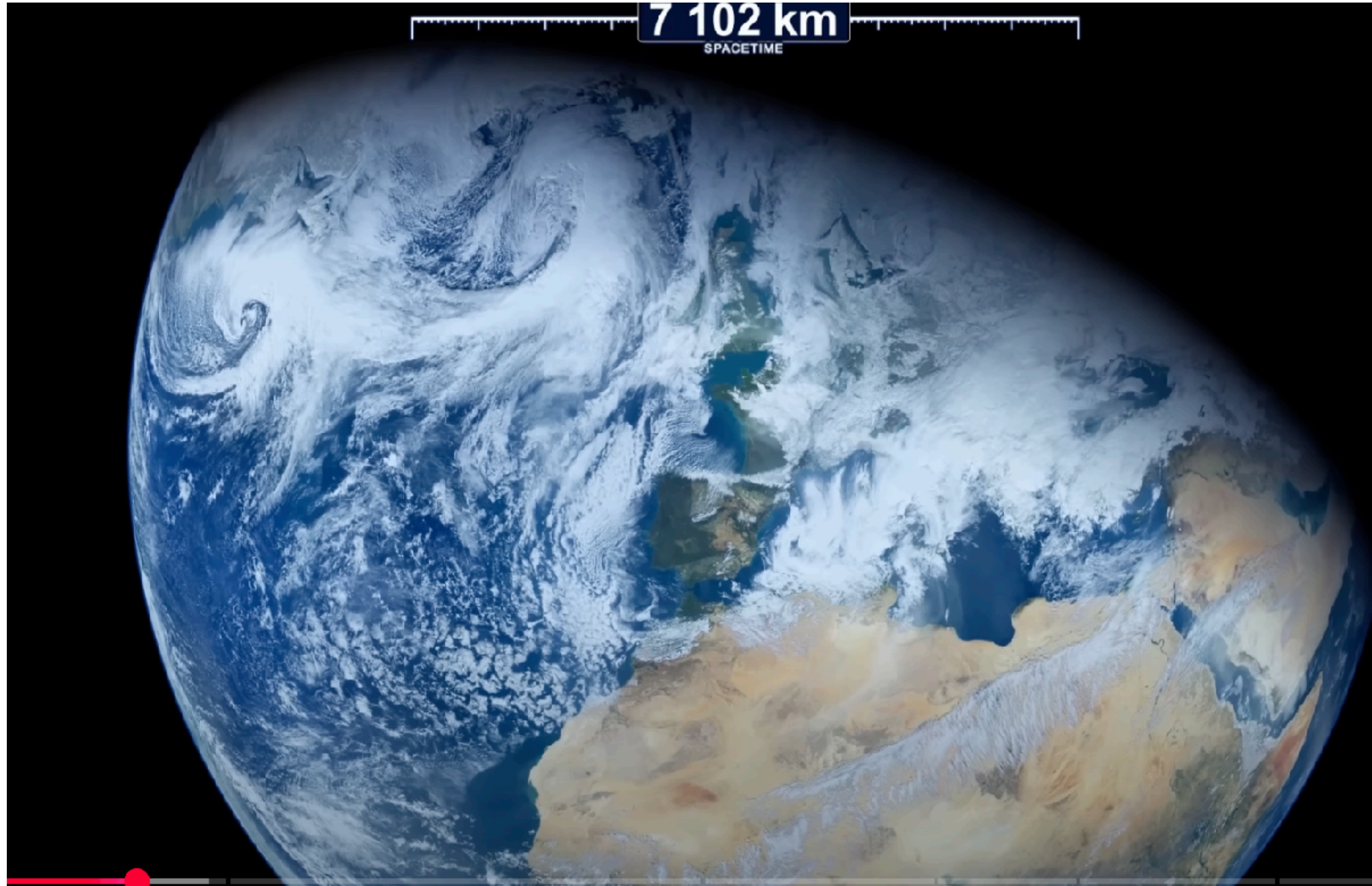


WHAT IS COSMOLOGY? NOT ABOUT YOUR HOUSE

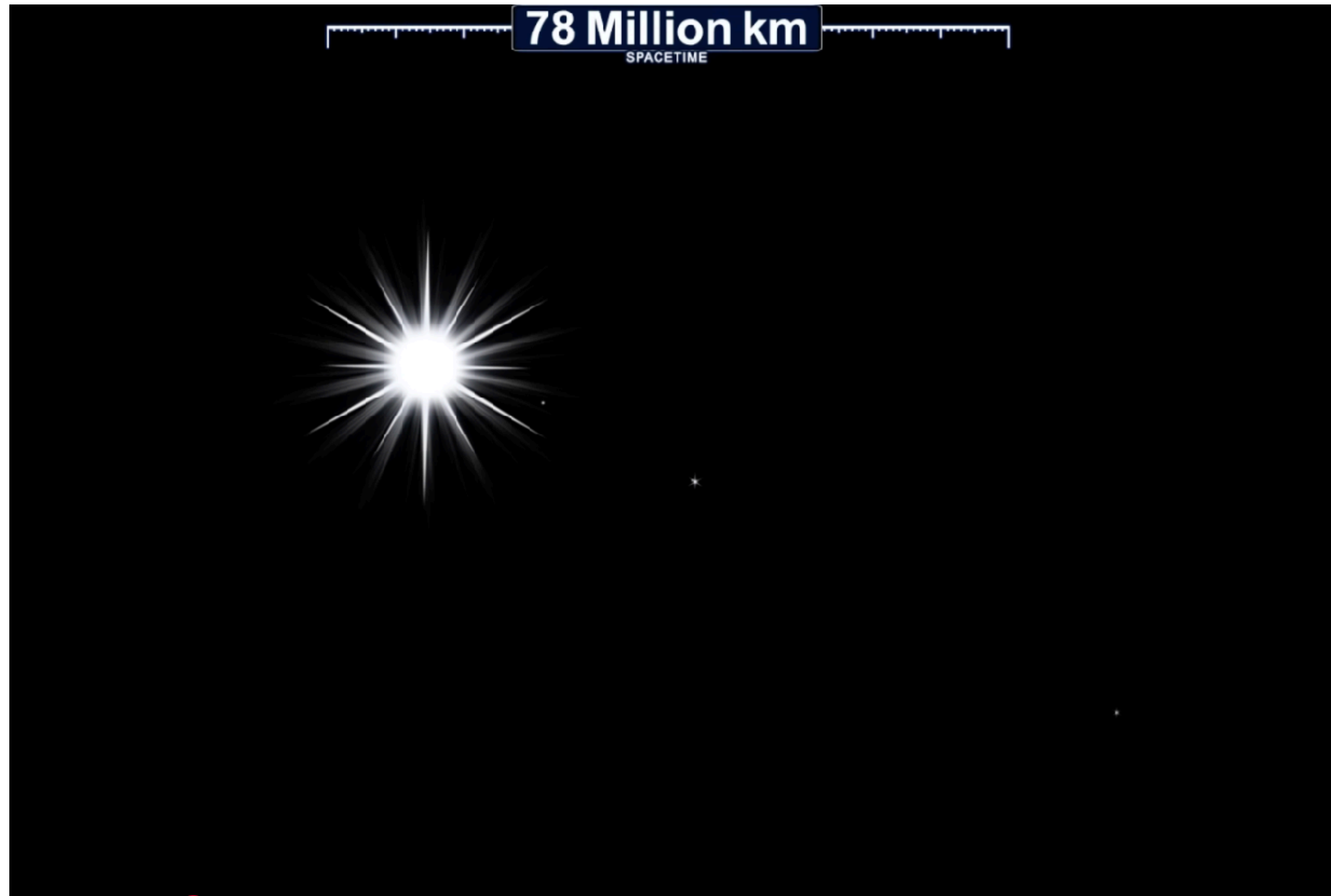


Credit: <https://www.youtube.com/watch?v=U9WOMerccoM&t=560s>

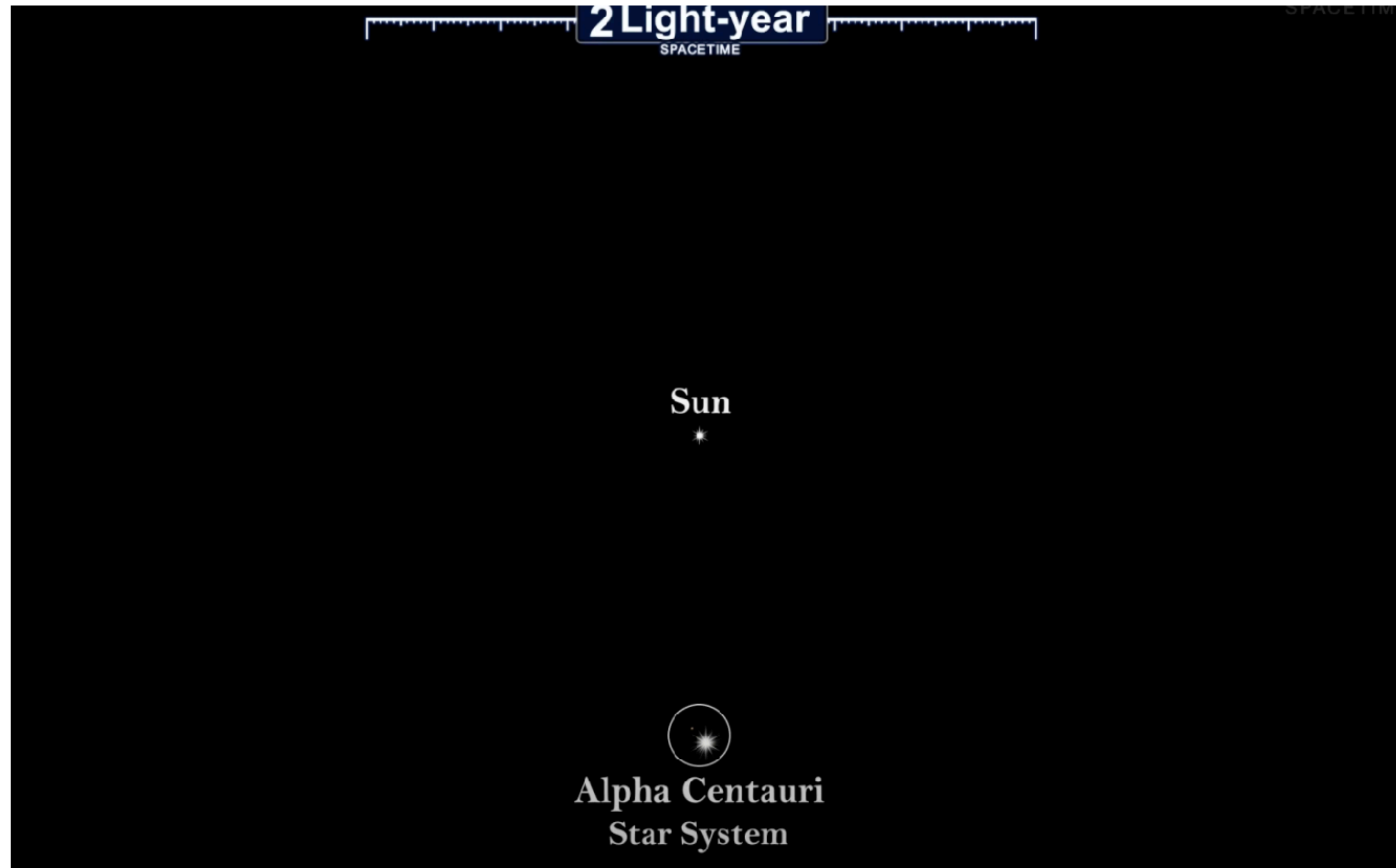
WHAT IS COSMOLOGY? NOT ABOUT YOUR RIDICULOUSLY SMALL PLANET



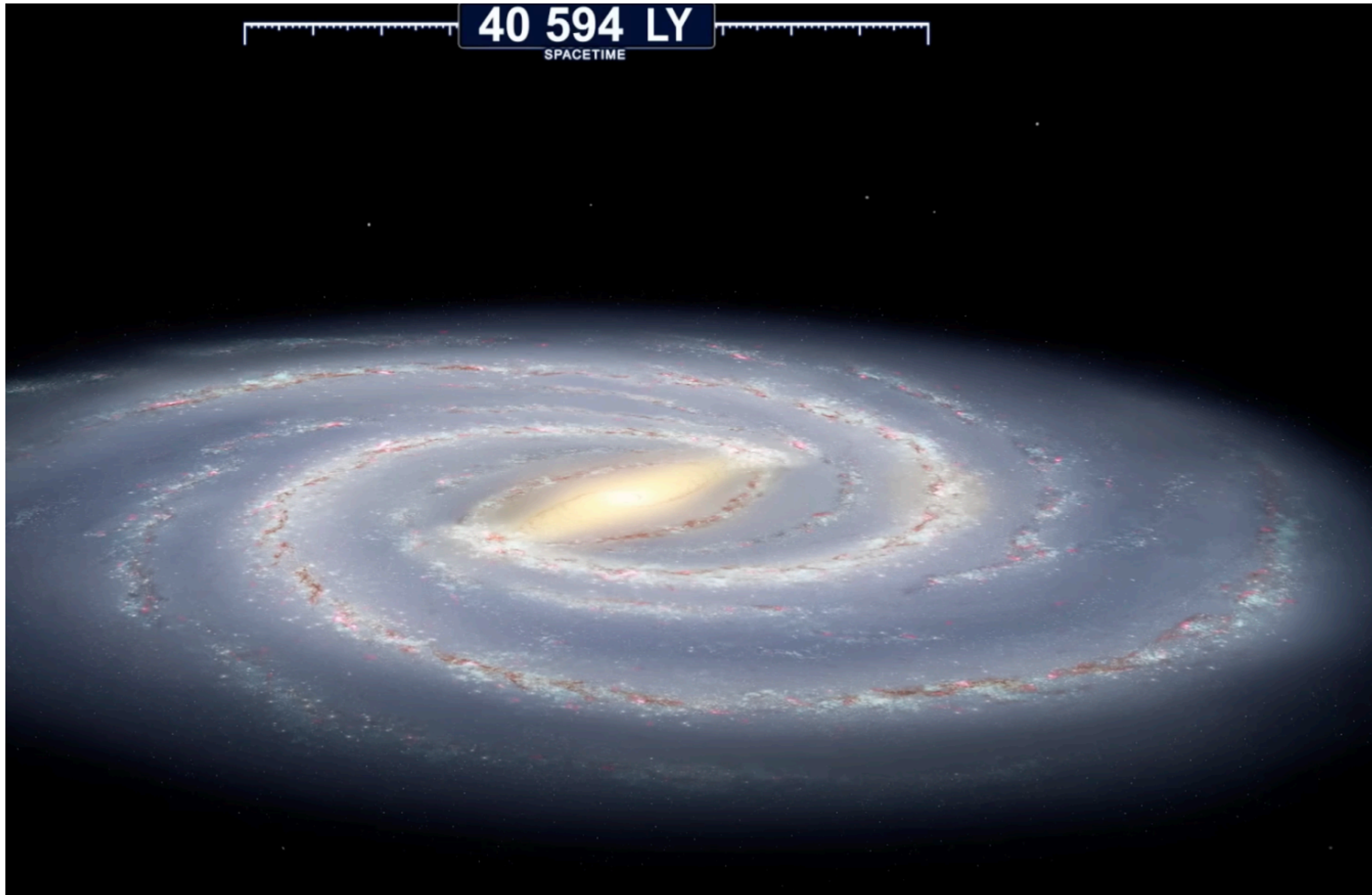
WHAT IS COSMOLOGY? NOT ABOUT YOUR INSIGNIFICANT SOLAR SYSTEM



WHAT IS COSMOLOGY? NOT ABOUT STARS



WHAT IS COSMOLOGY? NOT QUITE THERE YET



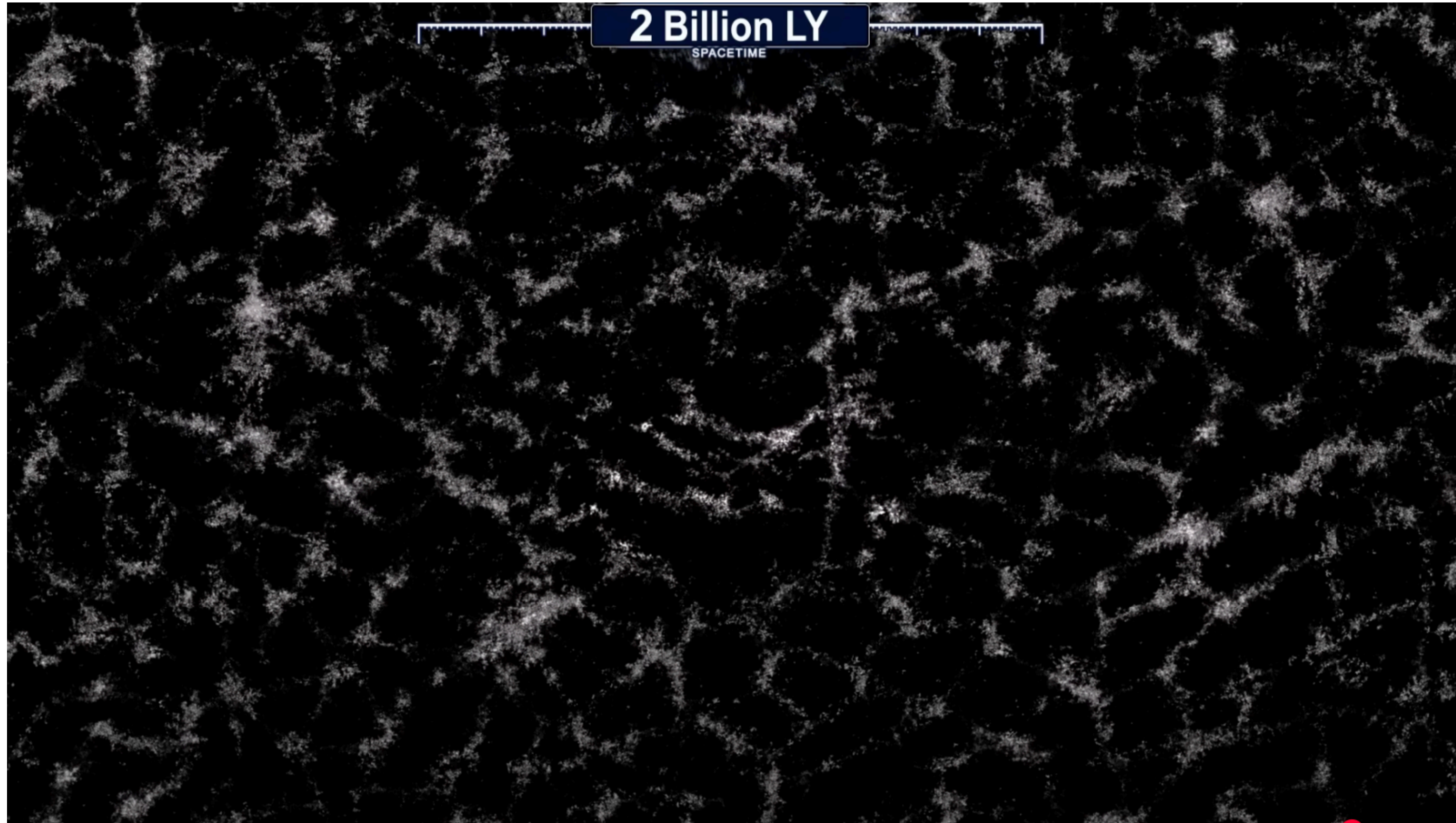
WHAT IS COSMOLOGY? NOW WE START TALKING



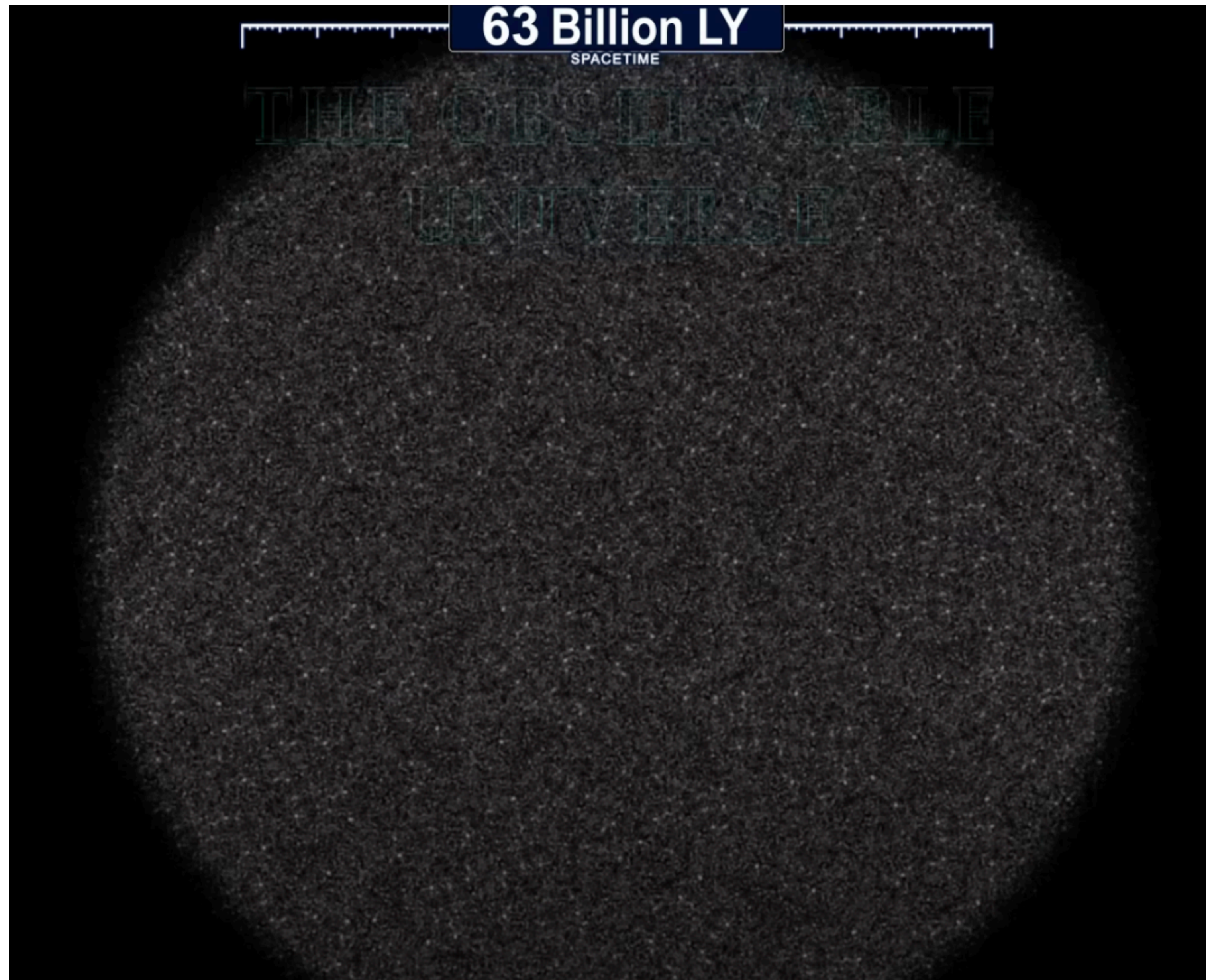
WHAT IS COSMOLOGY? NOW WE CAN THINK ABOUT COSMOLOGY



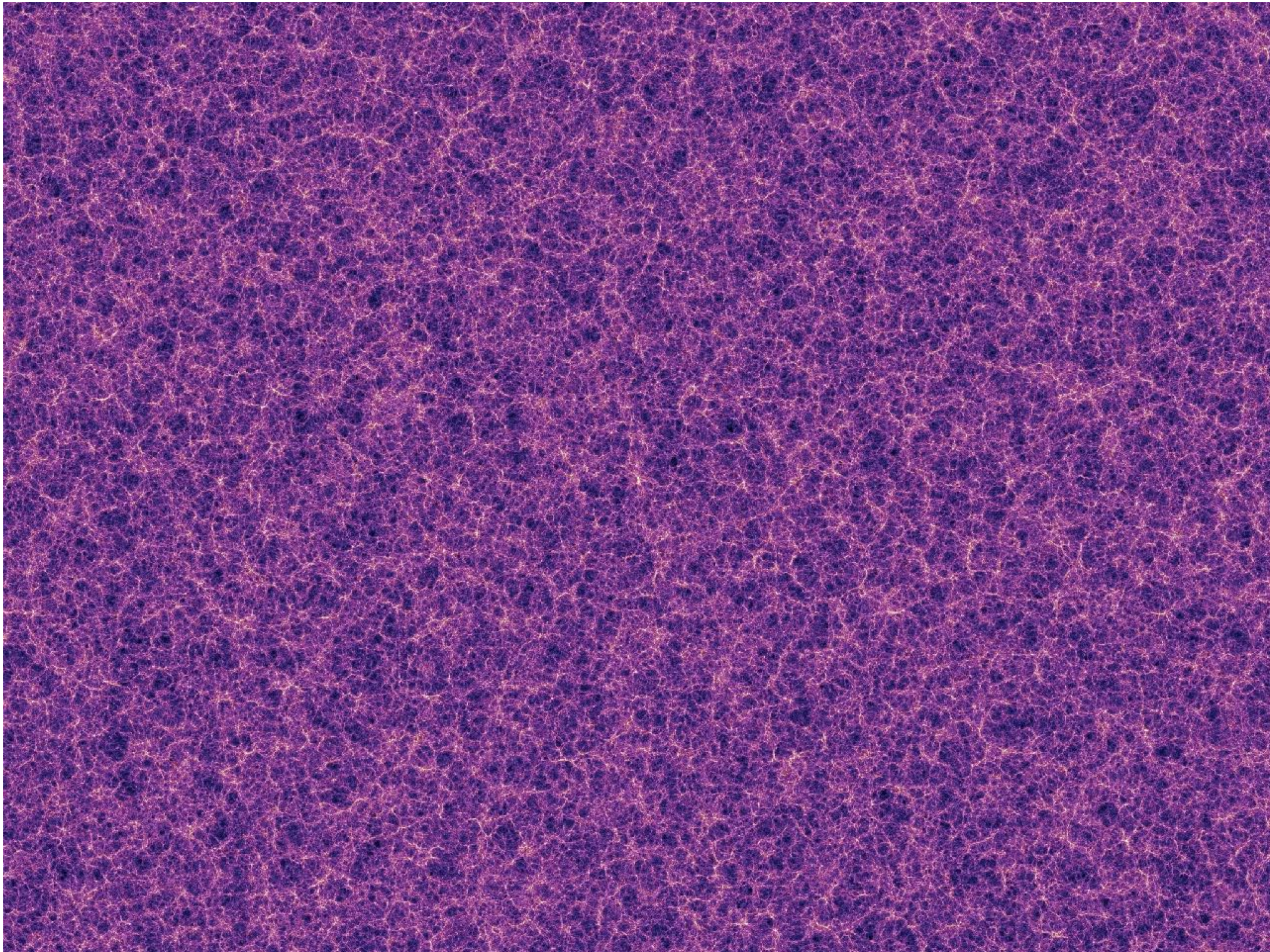
WHAT IS COSMOLOGY? NOW WE CAN ACTUALLY DO COSMOLOGY



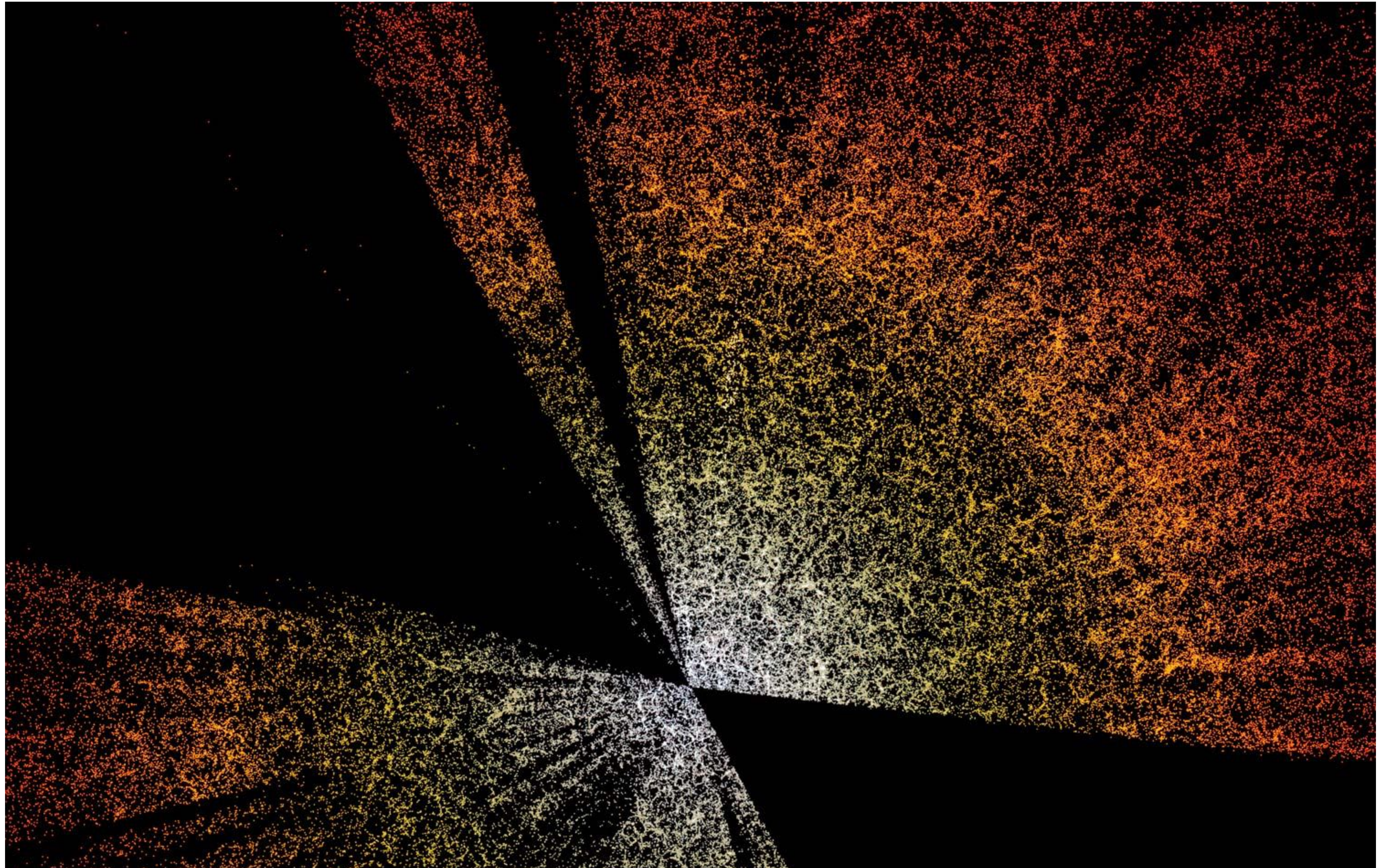
WHAT IS COSMOLOGY? EVERYTHING WE CAN EVER KNOW ABOUT COSMOLOGY



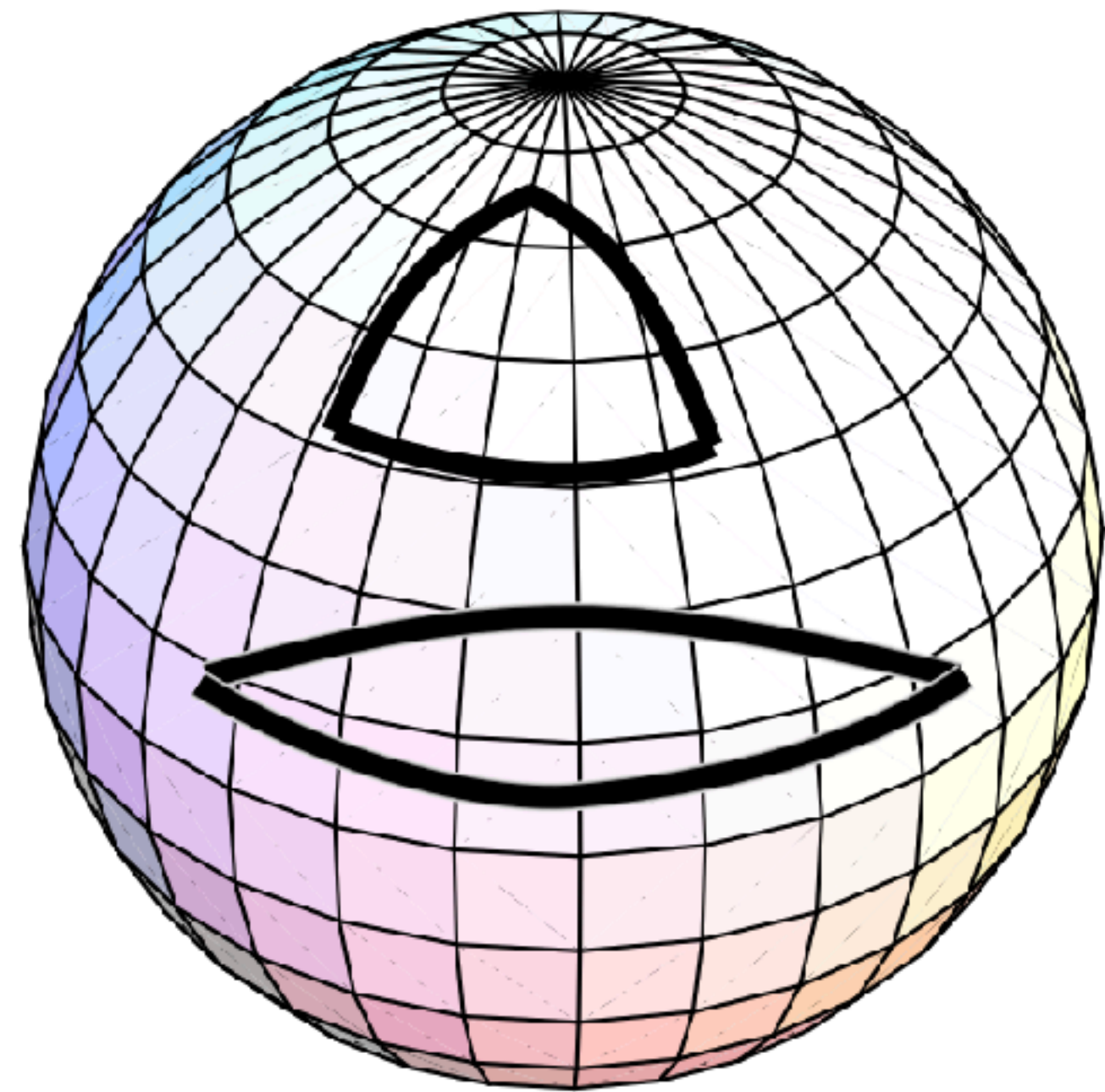
THE COSMOLOGICAL PRINCIPLE (CREDIT: MILLENNIUM SIMULATION)



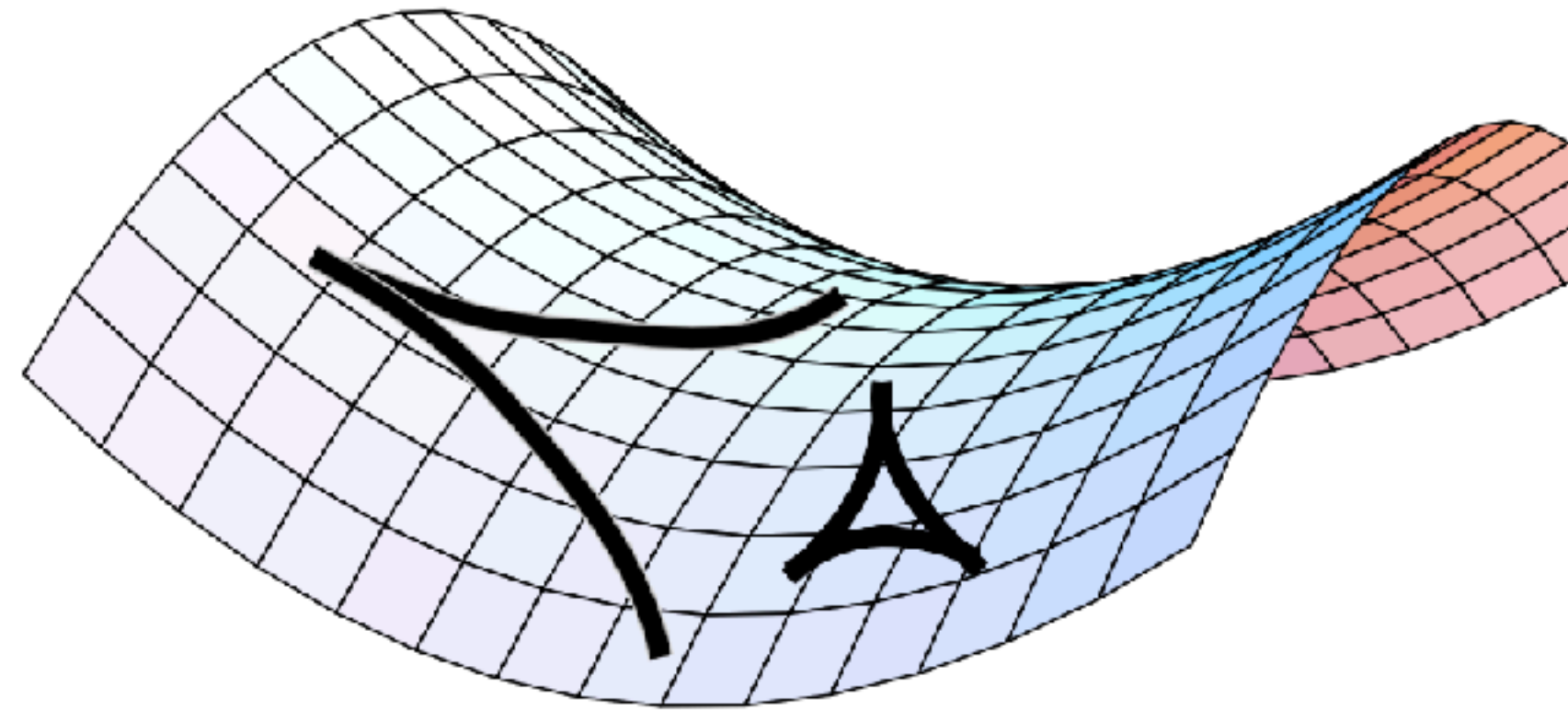
WHAT IS COSMOLOGY? GALAXY SURVEYS (CREDIT: DESI COLLABORATION)



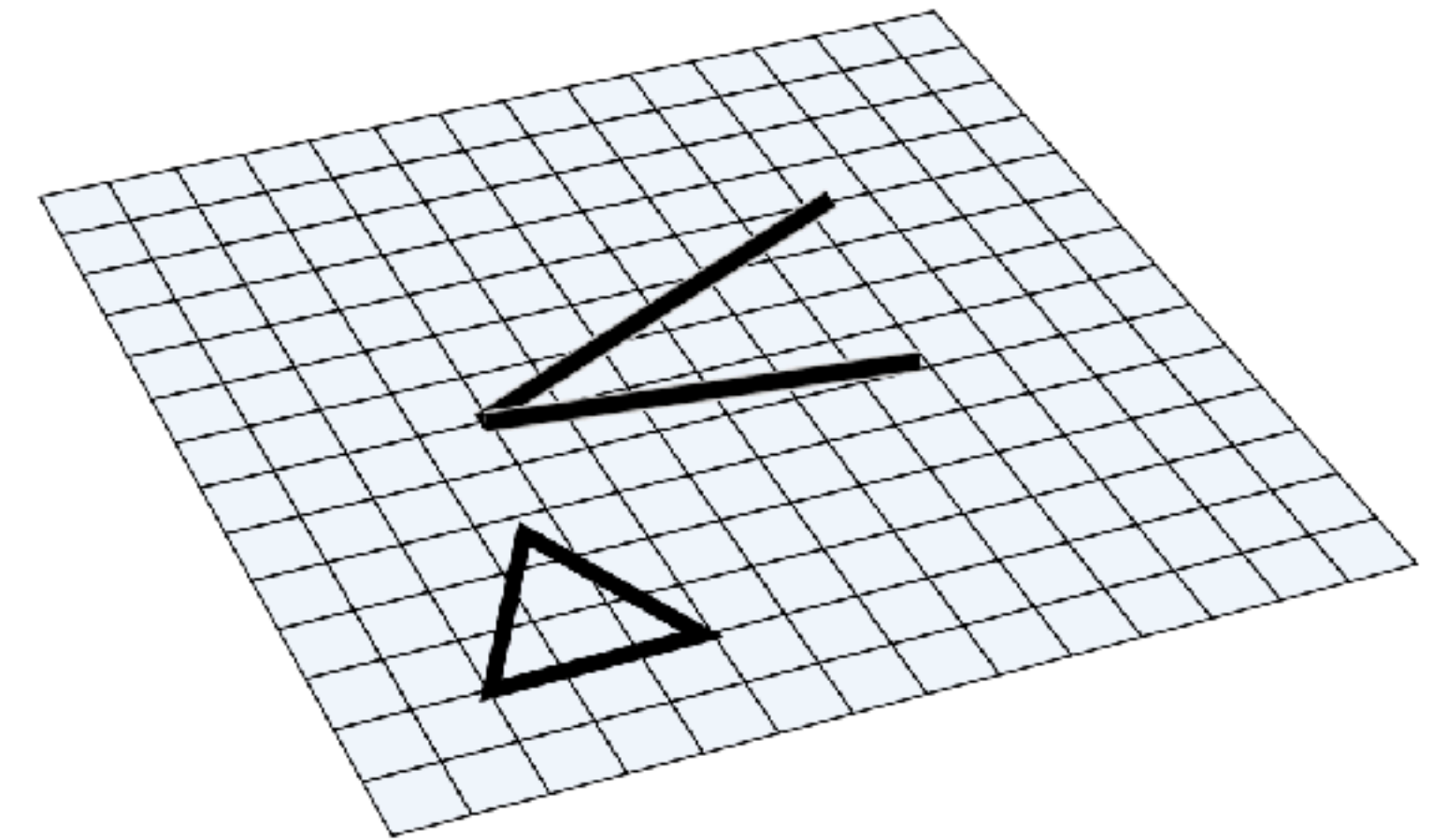
(POSSIBLE CURVATURES OF AN HOMOGENEOUS AND ISOTROPIC UNIVERSE)



Positive curvature

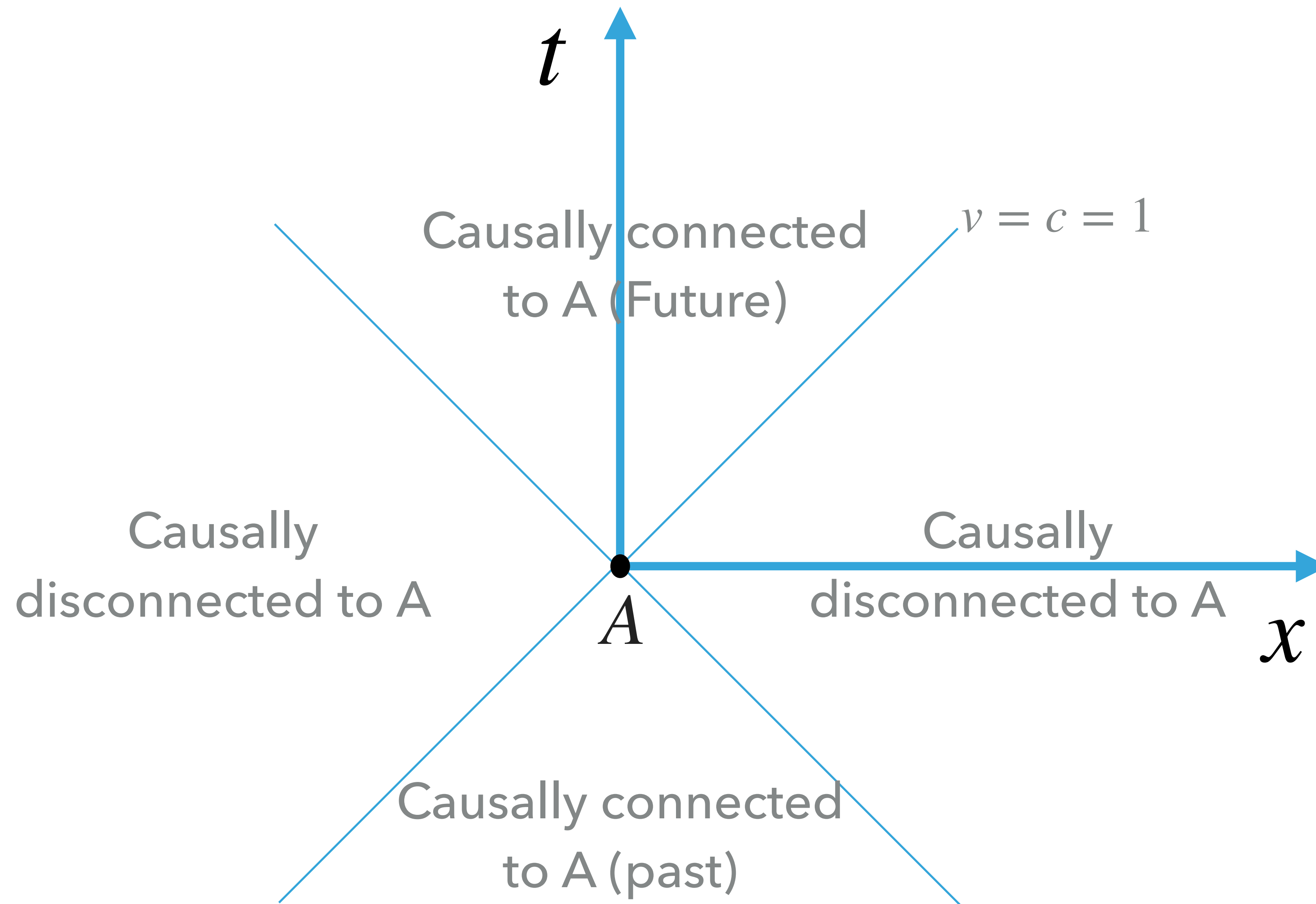


Negative curvature

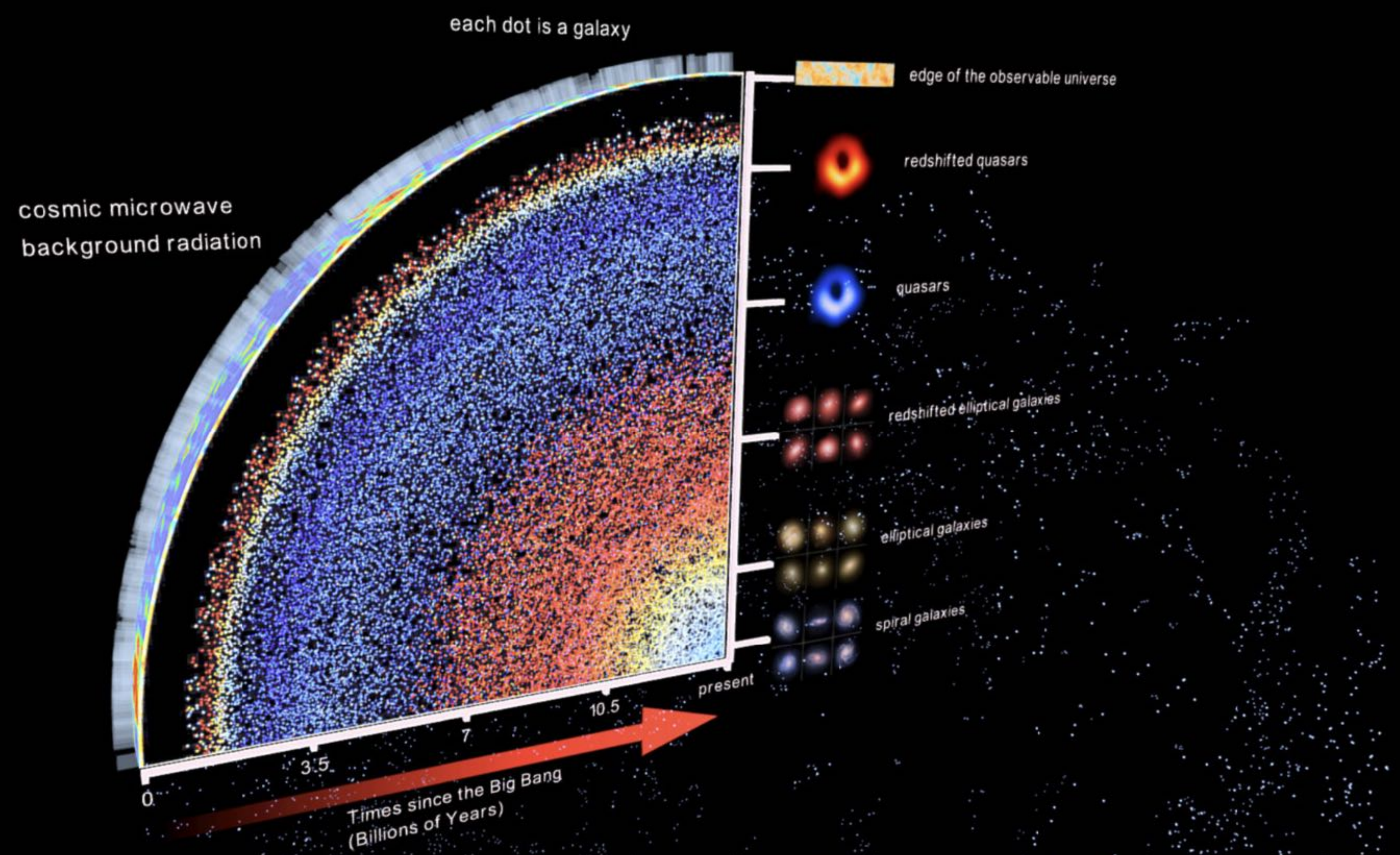


No curvature

WHAT IS THE UNIVERSE MADE OF – SPACETIME (EINSTEINIAN)



WHAT IS THE UNIVERSE MADE OF – SPACETIME (EINSTEINIAN) – CREDIT: SALVATORE ORLANDO

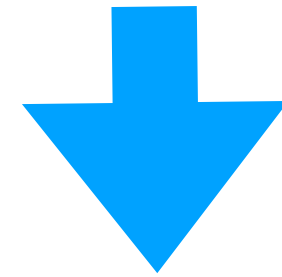


SUMMARY

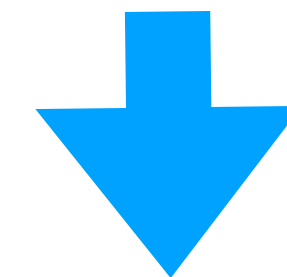
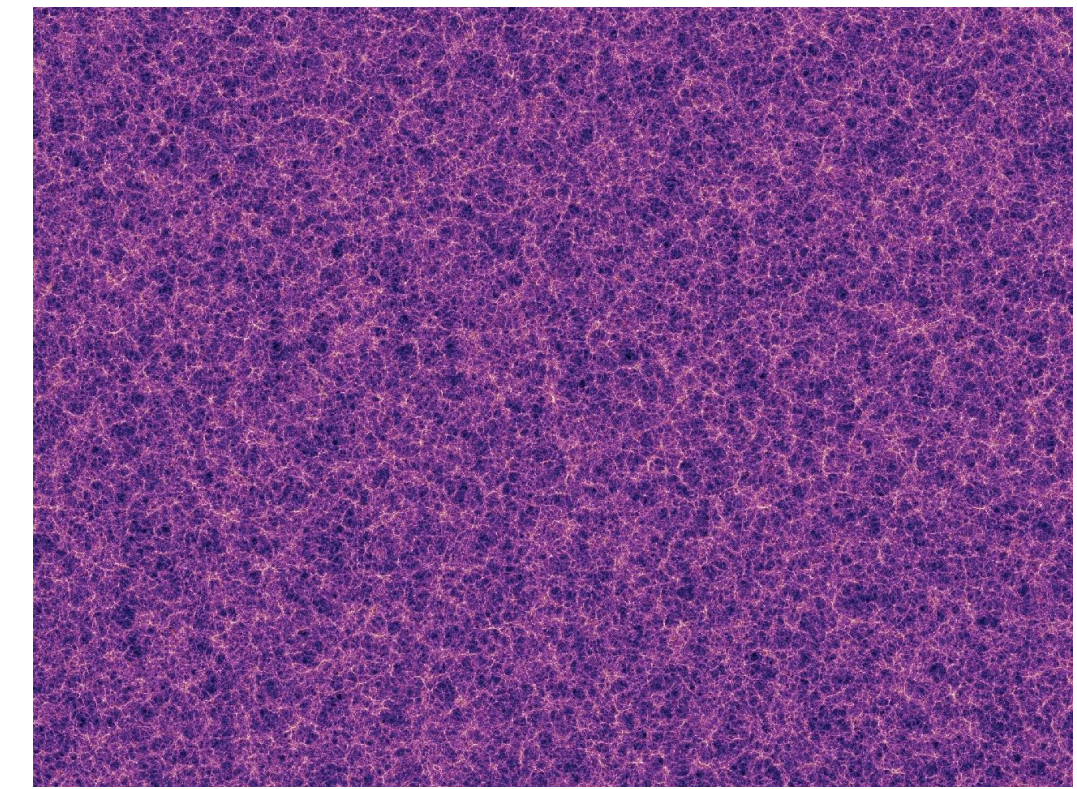
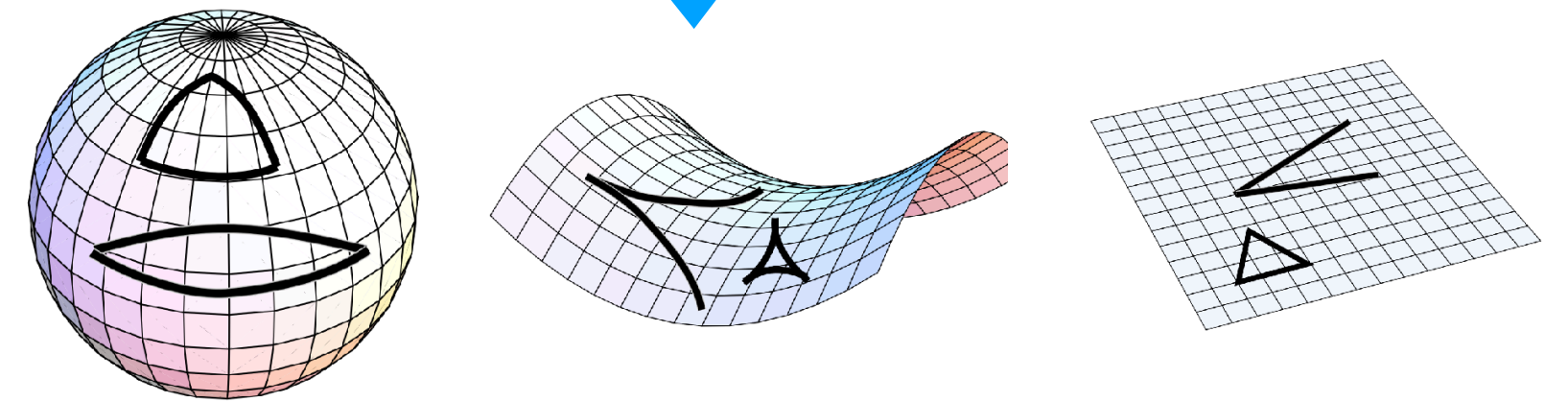
- Cosmology is the branch of physics that studies the Universe in its largest scales.
- On the largest scales, the Universe forms structures, as a result of gravitational collapse (and the initial conditions)
- On large scales, the Universe becomes Homogeneous and Isotropic, which means that, statistically, on a sufficient large scale, different patches of the Universe are indistinguishable.
- The Universe has a geometry. Since it is homogeneous and Isotropic, there are three possibilities: flat, closed (positive curvature), and hyperbolical (negative curvature).
- We can only observe a finite region of the Universe. This happens because there is a fundamental limit for how far a particle can travel (the speed of light).
- By looking to distant galaxies, we are also looking to the past, or to our past light cone.

RECAP

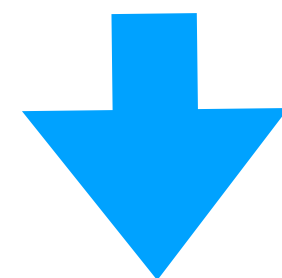
Homogeneous + Isotropy



3 possibilities for the geometry of the Universe

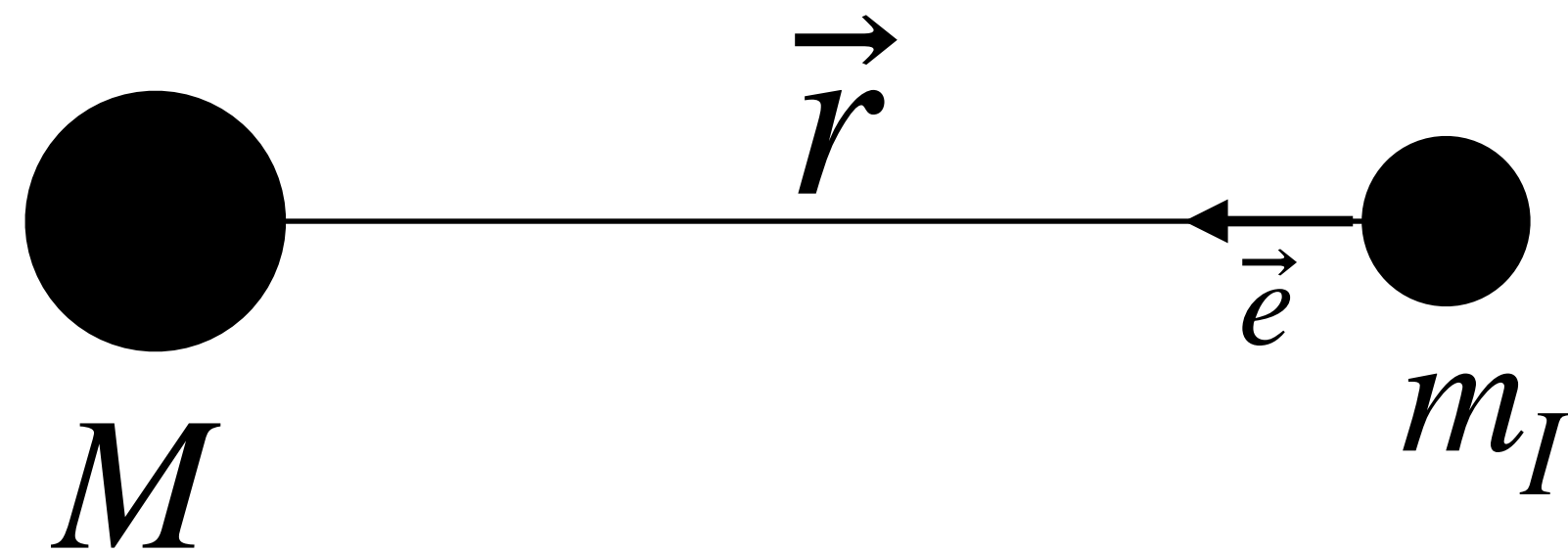


The speed of light is the fastest one allowed in the Universe



We only see a finite region of the Universe - our light cone

$$\vec{F} = m_I \vec{a} \quad \left(-\vec{\nabla} \phi(\vec{x}) = m \frac{d^2 \vec{x}(t)}{dt^2} \right)$$



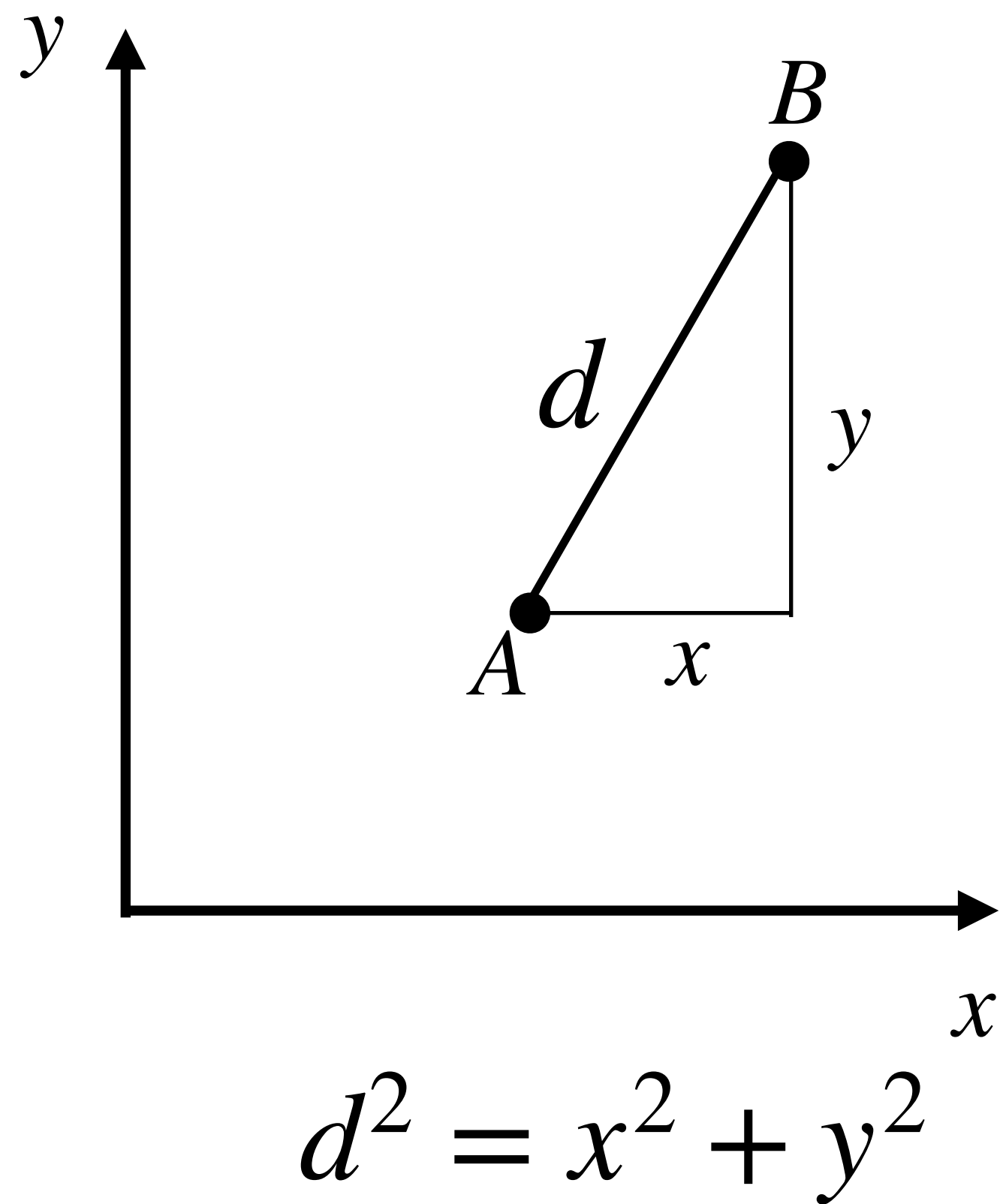
$$|\vec{e}| = 1$$

$$\vec{F} = G \frac{m_G M}{r^2} \vec{e} \Rightarrow m_I \vec{a} = G \frac{m_G M}{r^2} \vec{e}$$

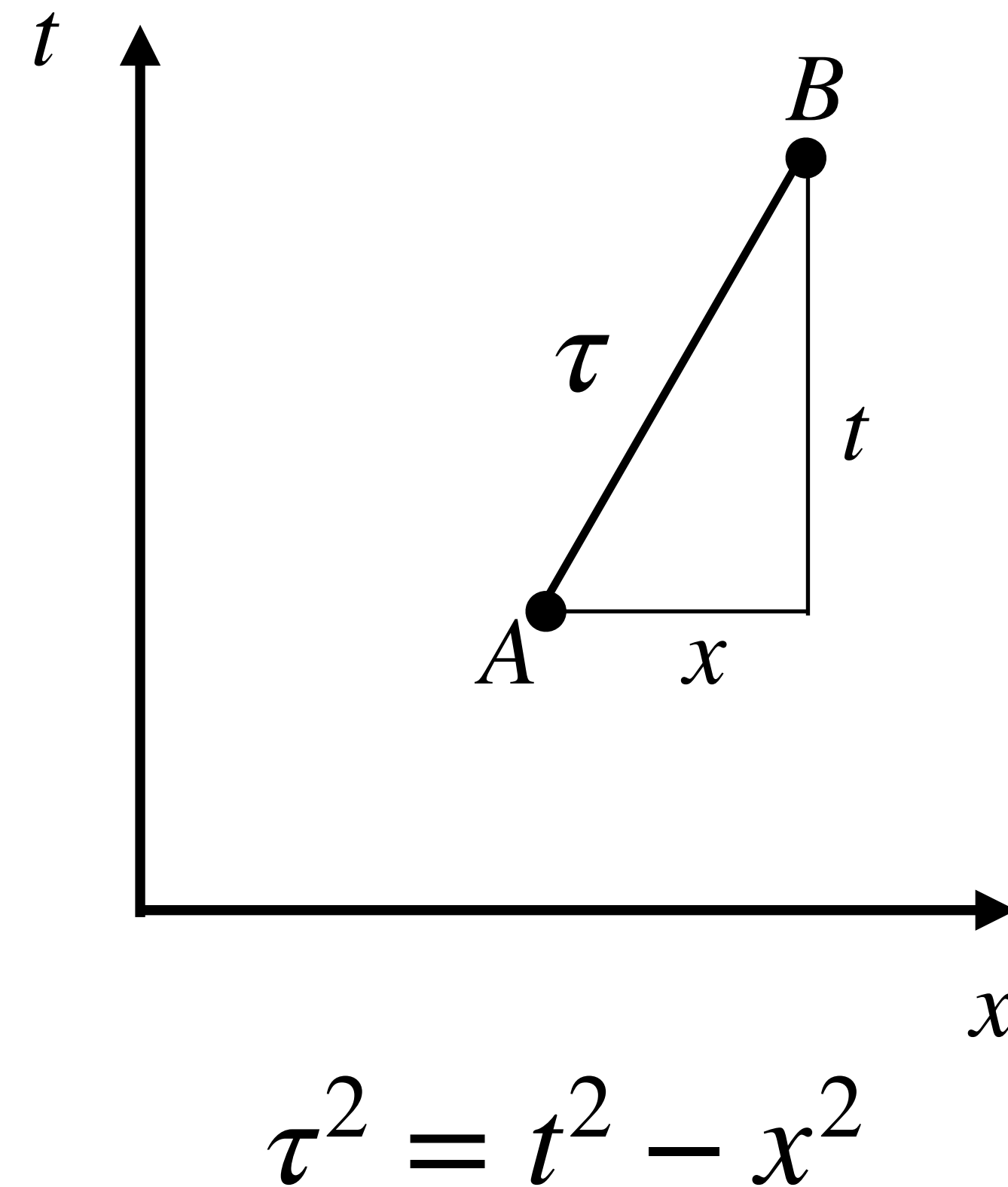


SPECIAL RELATIVITY (1905)

In Euclidean geometry, we calculate **distances** with Pitagora's theorem



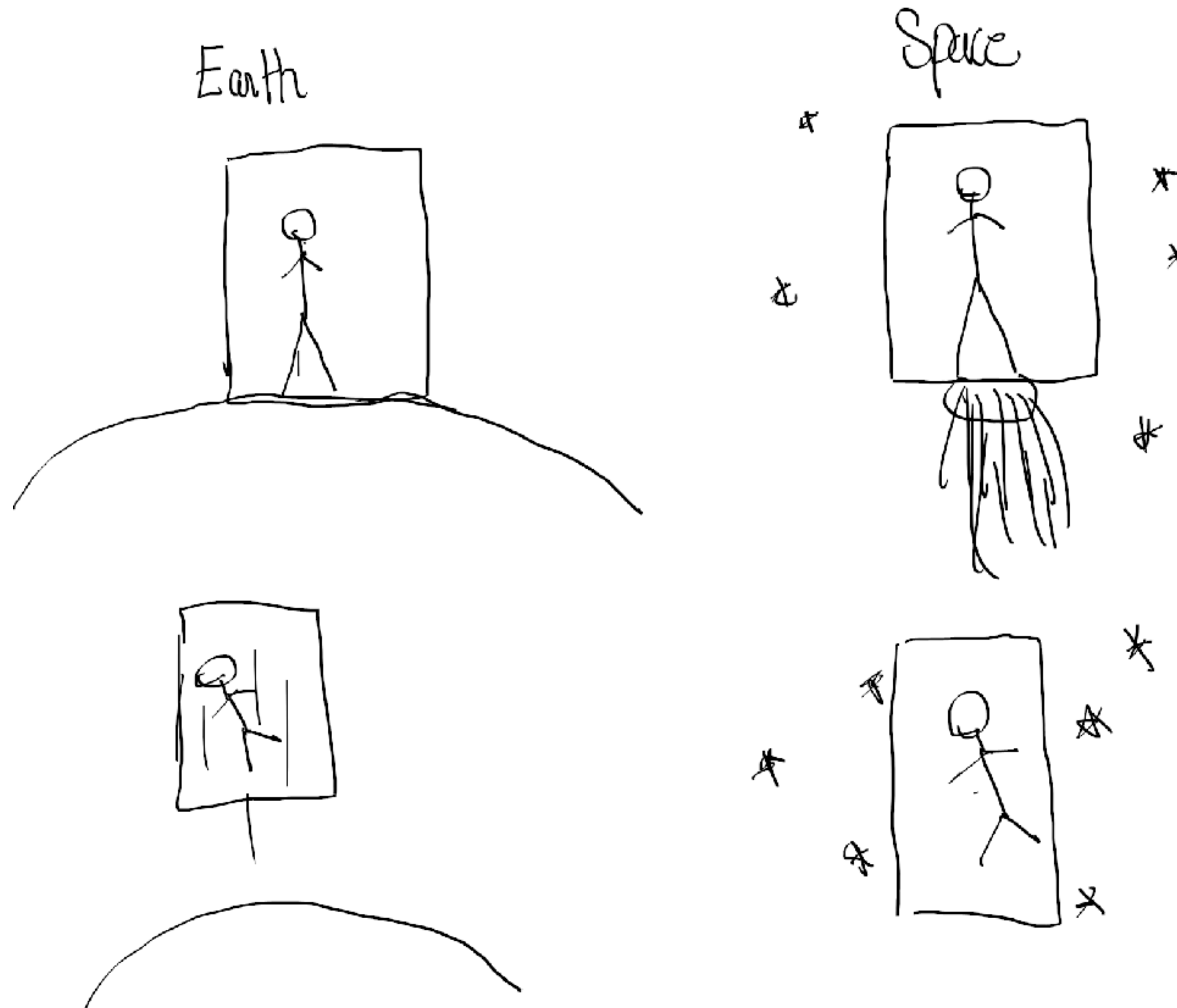
In Minkowskian geometry, we calculate **distances** with a modified version of Pitagora's theorem



The speed of light is invariant!

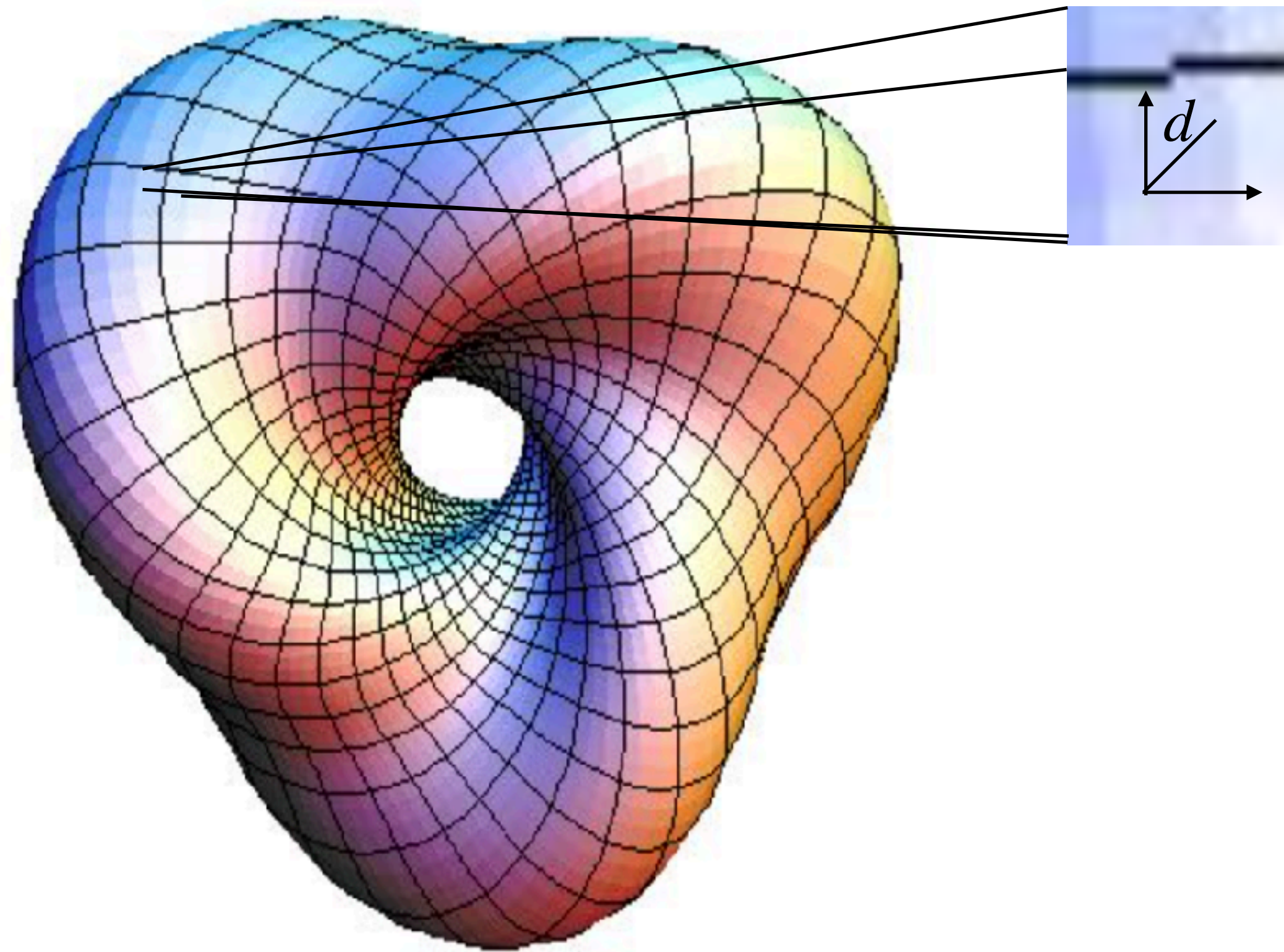
TOWARDS GENERAL RELATIVITY – THE EQUIVALENCE PRINCIPLE

Einstein's happiest thought:



TOWARDS GENERAL RELATIVITY – RIEMANNIANN GEOMETRY

Riemanniann manifold



Bernhard Riemann (1826 - 1866)

<https://people.math.wisc.edu/~gmaribef/761.html>

TOWARDS GENERAL RELATIVITY – RIEMANNIANN GEOMETRY

How to express a distance (interval) in an arbitrary Riemannian manifold?

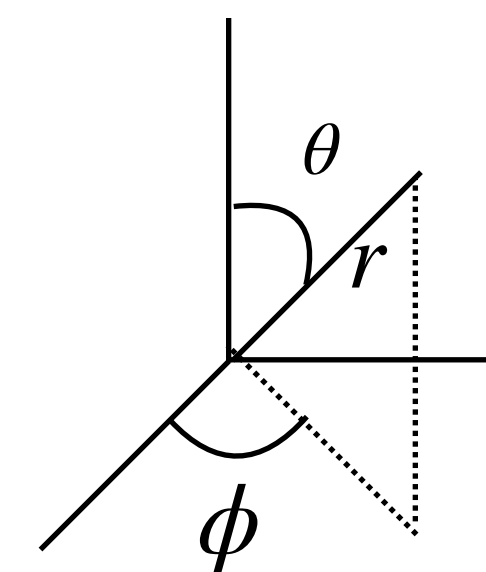


Distance² = some coefficient x some coordinate x same or another coordinate

Sum for each pair of coordinates.

In the case of Euclidean space:

$$d^2 = x^2 + y^2 + z^2 \quad \text{coordinates } (x, y, z) \quad \text{coefficients } (1, 1, 1)$$



In spherical coordinates:

$$d^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2(\theta) d\phi^2 \quad \text{coordinates } (r, \theta, \phi) \quad \text{coefficients } (1, r^2, r^2 \sin^2(\theta))$$

In a general spacetime:

$$ds^2 = \text{interval}^2 = A dt^2 + B dt dx + C dt dy + D dt dz + E dx dt + F dx^2 + G dx dy + \dots$$

TOWARDS GENERAL RELATIVITY – RIEMANNIANN GEOMETRY

Let's simplify the notation:

$$\{t, x, y, z\} = \{x^0, x^1, x^2, x^3\} = x^\mu$$

Let's simplify the notation by defining the **metric tensor** $g_{\mu\nu}$

$$\begin{aligned} ds^2 = & g_{00}(dx^0)^2 + g_{01}dx^0dx^1 + g_{02}dx^0dx^2 + g_{03}dx^0dx^3 \\ & g_{10}dx^1dx^0 + g_{11}(dx^1)^2 + g_{12}dx^1dx^2 + g_{13}dx^1dx^3 \\ & g_{20}dx^2dx^0 + g_{21}dx^2dx^1 + g_{22}(dx^2)^2 + g_{23}dx^2dx^3 \\ & g_{30}dx^3dx^0 + g_{31}dx^3dx^1 + g_{32}dx^3dx^2 + g_{33}(dx^3)^2 \end{aligned}$$

$$ds^2 = \sum_{\mu\nu} g_{\mu\nu} dx^\mu dx^\nu$$



TOWARDS GENERAL RELATIVITY – RIEMANNIANN GEOMETRY

E.g. for Minkowski spacetime:

$$\tau^2 = t^2 - x^2 - y^2 - z^2$$

$$g_{\mu\nu} = \begin{pmatrix} g_{tt} & g_{tx} & g_{ty} & g_{tz} \\ g_{xt} & g_{xx} & g_{xy} & g_{xz} \\ g_{yt} & g_{yx} & g_{yy} & g_{yz} \\ g_{zt} & g_{zx} & g_{zy} & g_{zz} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

The metric tensor $g_{\mu\nu}$ contains all the information about the geometry of spacetime

TOWARDS GENERAL RELATIVITY – RIEMANNIANN GEOMETRY

E.g. for Minkowski spacetime:

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2$$

Rate at which time flows Time and space twist into each other

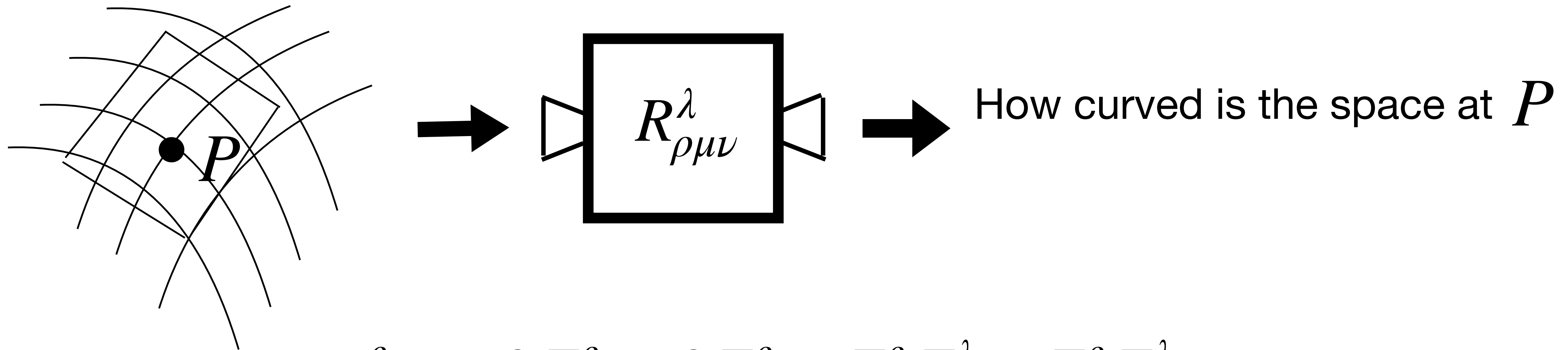
$$g_{\mu\nu} = \begin{pmatrix} \boxed{g_{tt}} & \boxed{g_{tx} \quad g_{ty} \quad g_{tz}} \\ \boxed{g_{xt}} & \boxed{g_{xx} \quad g_{xy} \quad g_{xz}} \\ \boxed{g_{yt}} & \boxed{g_{yx} \quad g_{yy} \quad g_{yz}} \\ \boxed{g_{zt}} & \boxed{g_{zx} \quad g_{zy} \quad g_{zz}} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Spatial distances

The metric tensor $g_{\mu\nu}$ contains all the information about the geometry of spacetime

TOWARDS GENERAL RELATIVITY – RIEMANNIANN GEOMETRY

The Riemann tensor $R^\lambda_{\rho\mu\nu}$ is defined in each point of the manifold and measures curvature.



$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$$

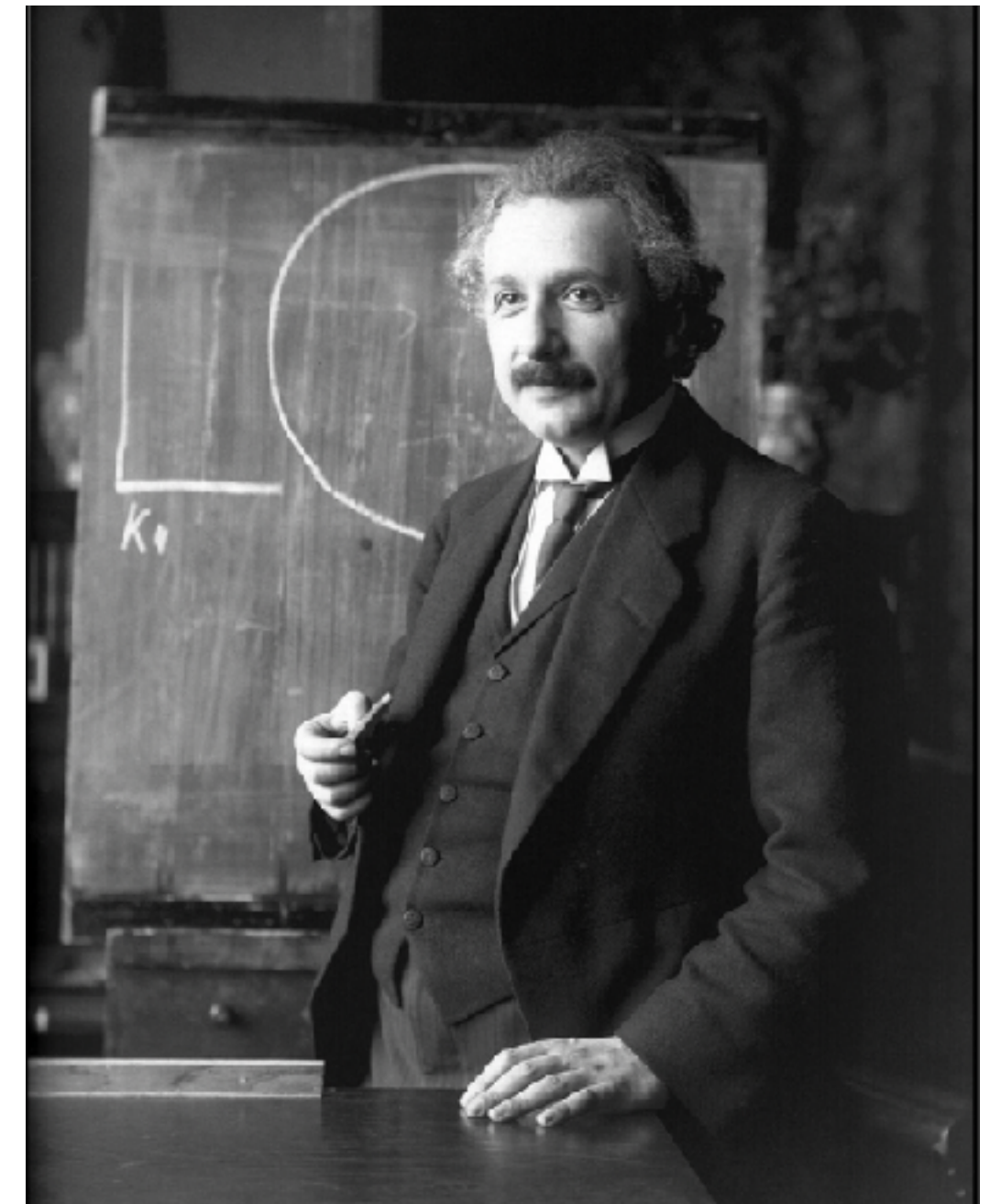
$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} \left(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu} \right)$$

TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

How to use all of this to get a theory of gravity? (Einstein)

We need something like

$$\vec{a} = \frac{GM}{r^2} \vec{e}$$



Albert Einstein (1879 - 1955)

TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

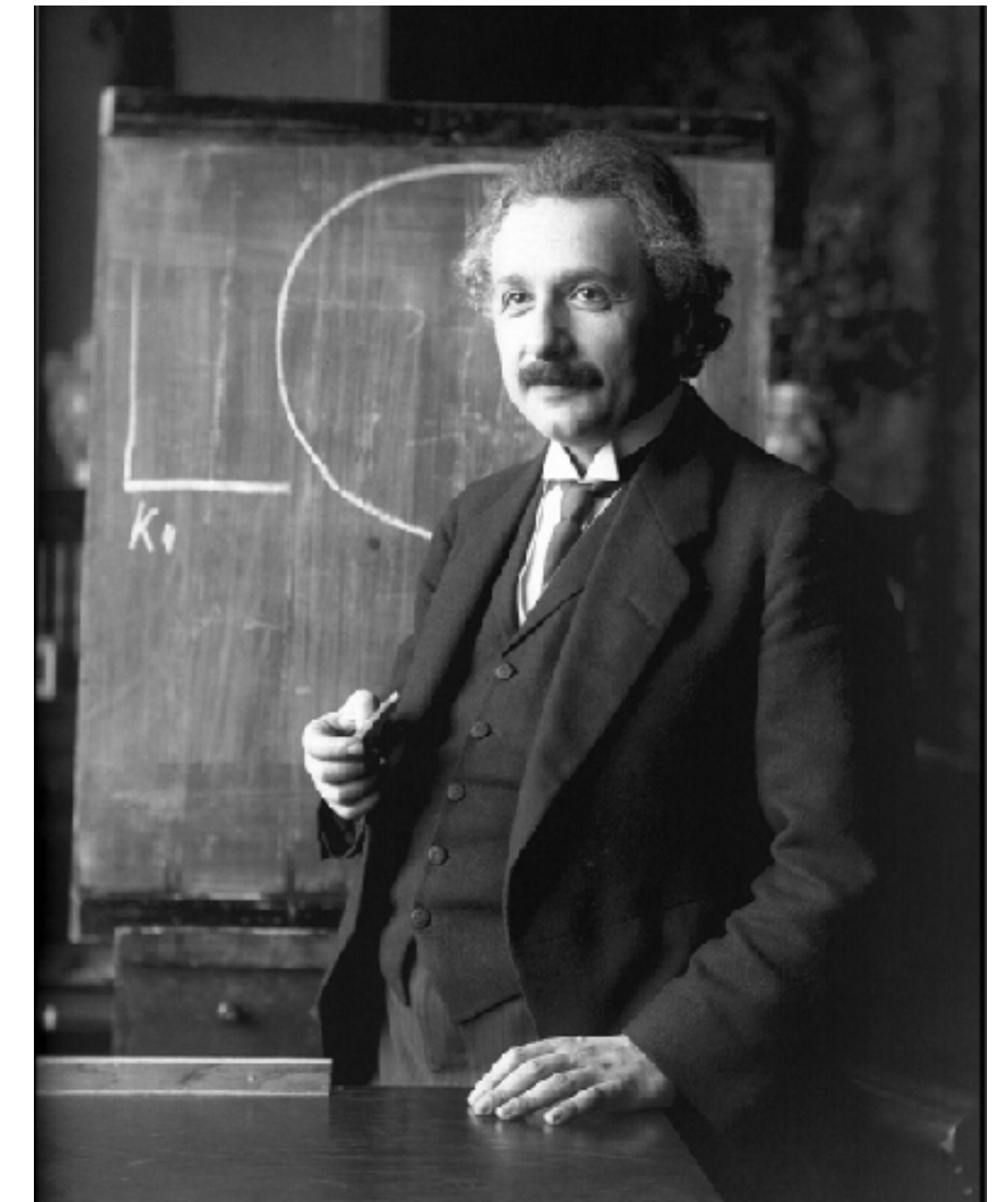
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Need to
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Need to replace
by how matter/
energy
($E = mc^2$) is
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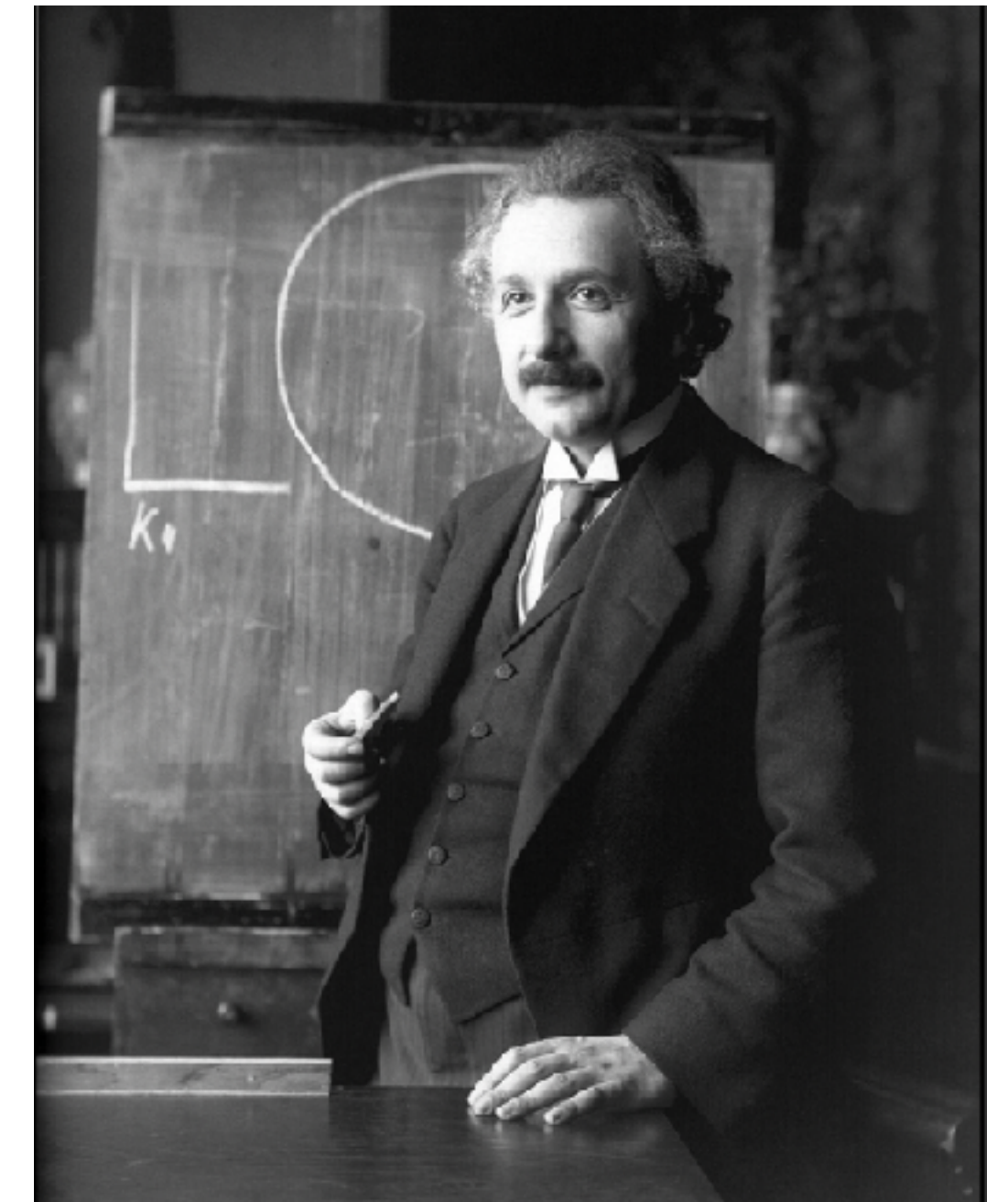
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We need a tensor for describing matter/energy:

$$T_{\mu\nu} = \begin{pmatrix} T_{tt} & T_{tx} & T_{ty} & T_{tz} \\ T_{xt} & T_{xx} & T_{xy} & T_{xz} \\ T_{yt} & T_{yx} & T_{yy} & T_{yz} \\ T_{zt} & T_{zx} & T_{zy} & T_{zz} \end{pmatrix}$$



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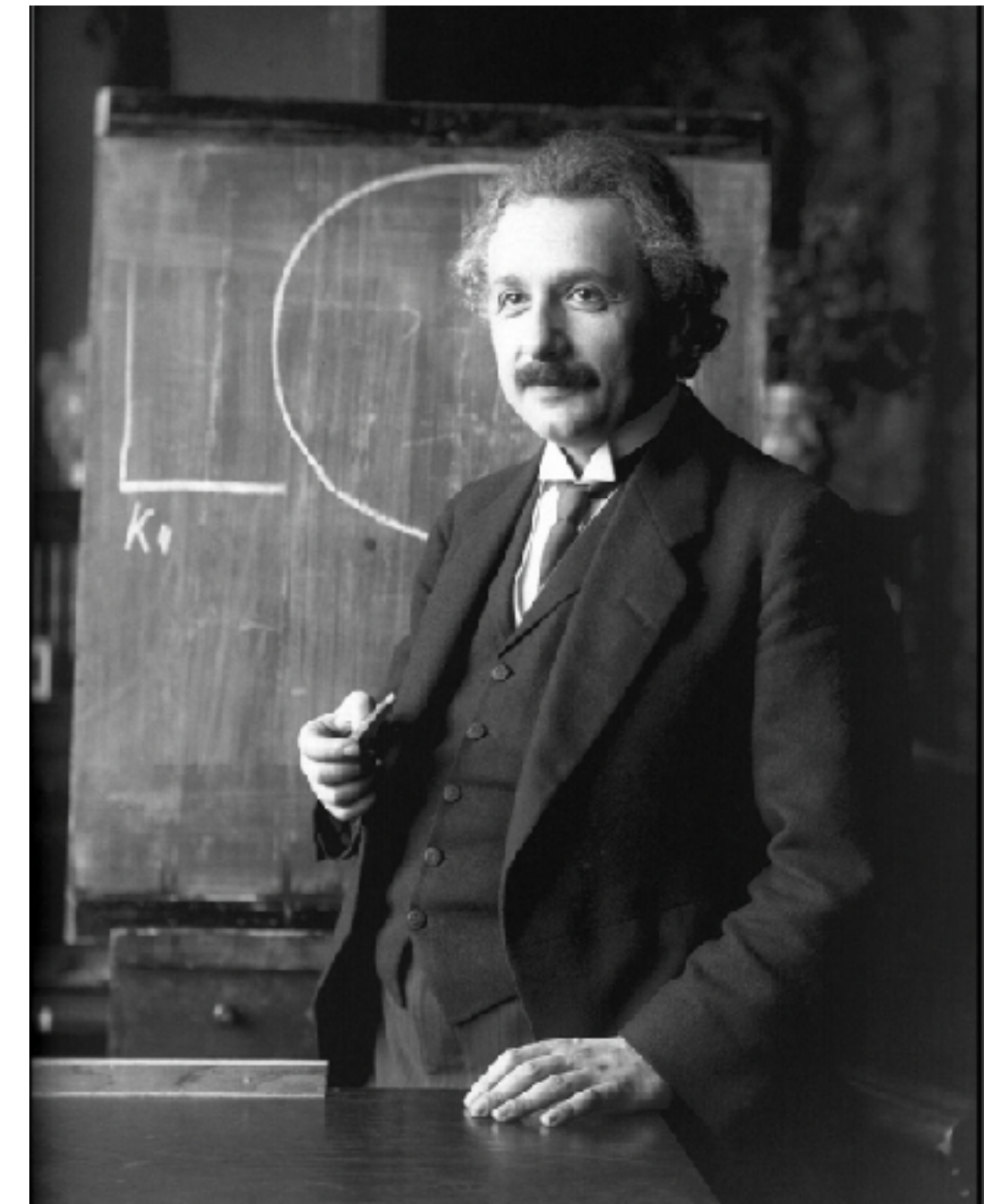
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We need a tensor for describing matter/energy:

Energy

$$T_{\mu\nu} = \begin{pmatrix} T_{tt} & T_{tx} & T_{ty} & T_{tz} \\ T_{xt} & T_{xx} & T_{xy} & T_{xz} \\ T_{yt} & T_{yx} & T_{yy} & T_{yz} \\ T_{zt} & T_{zx} & T_{zy} & T_{zz} \end{pmatrix}$$



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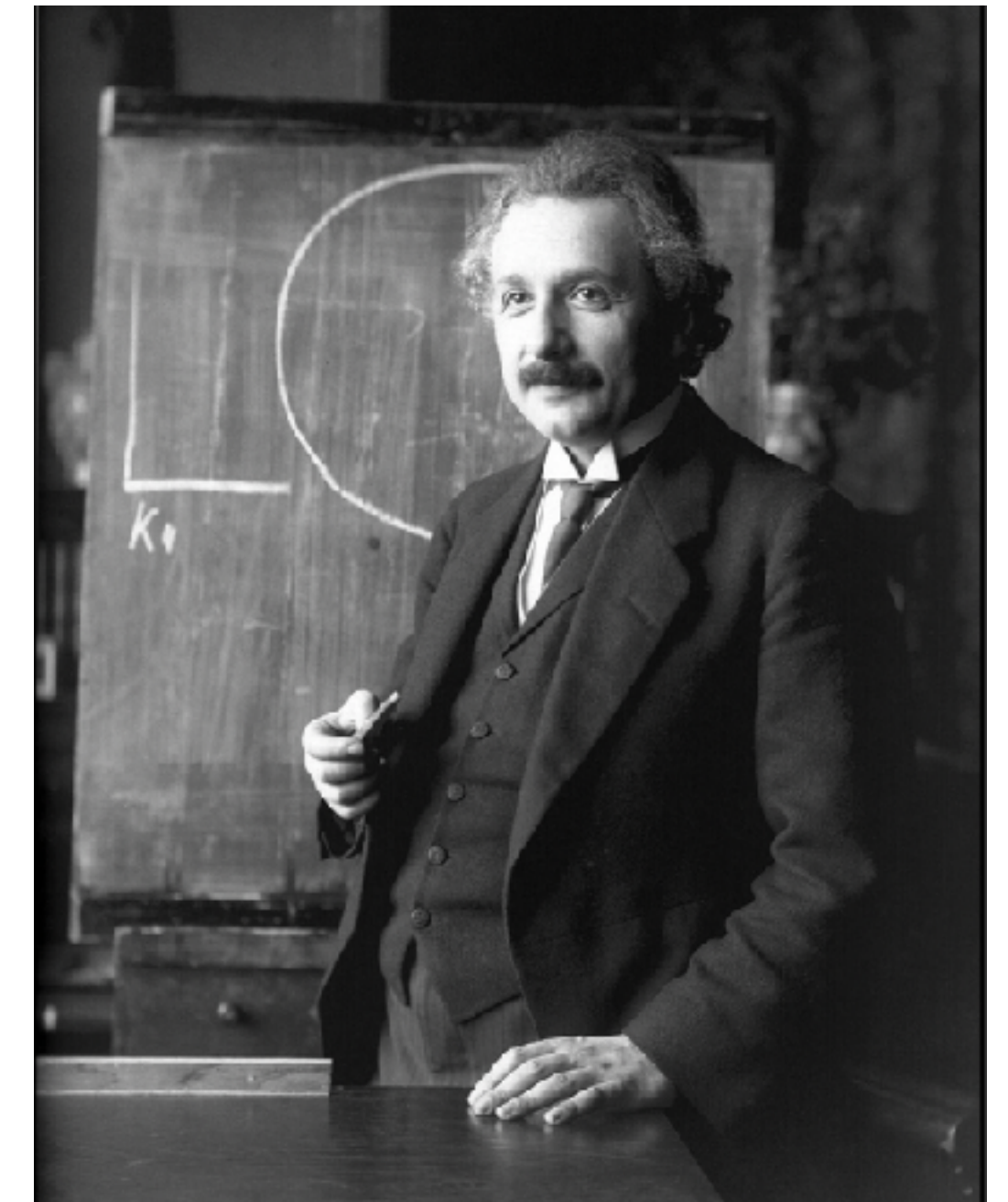
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Pressure along the x, y and z directions



Albert Einstein (1879 - 1955)

TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

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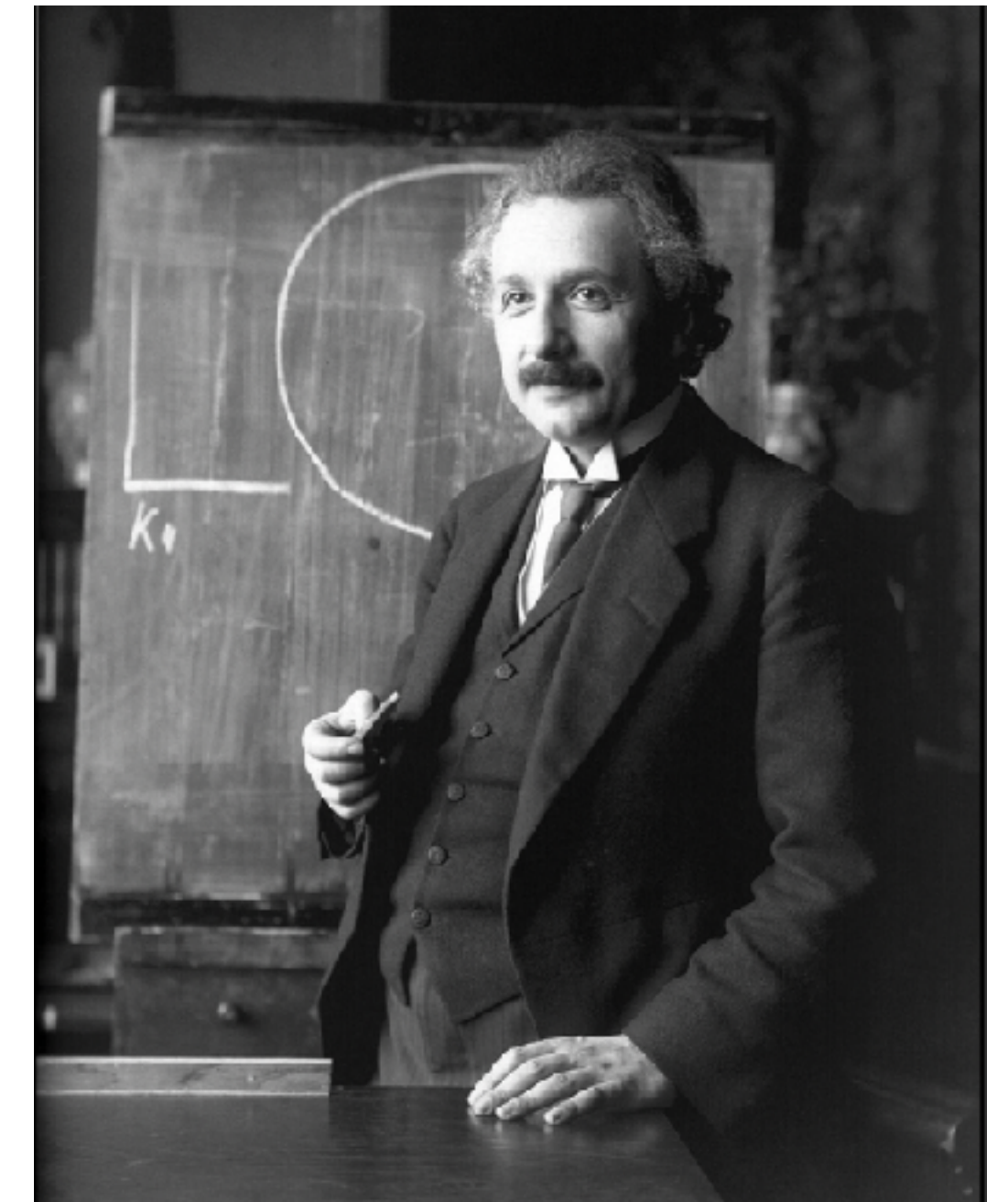
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We need a tensor for describing matter/energy:

Energy flux along the x, y and z directions

$$T_{\mu\nu} = \begin{pmatrix} T_{tt} & T_{tx} & T_{ty} & T_{tz} \\ T_{xt} & T_{xx} & T_{xy} & T_{xz} \\ T_{yt} & T_{yx} & T_{yy} & T_{yz} \\ T_{zt} & T_{zx} & T_{zy} & T_{zz} \end{pmatrix}$$



Albert Einstein (1879 - 1955)

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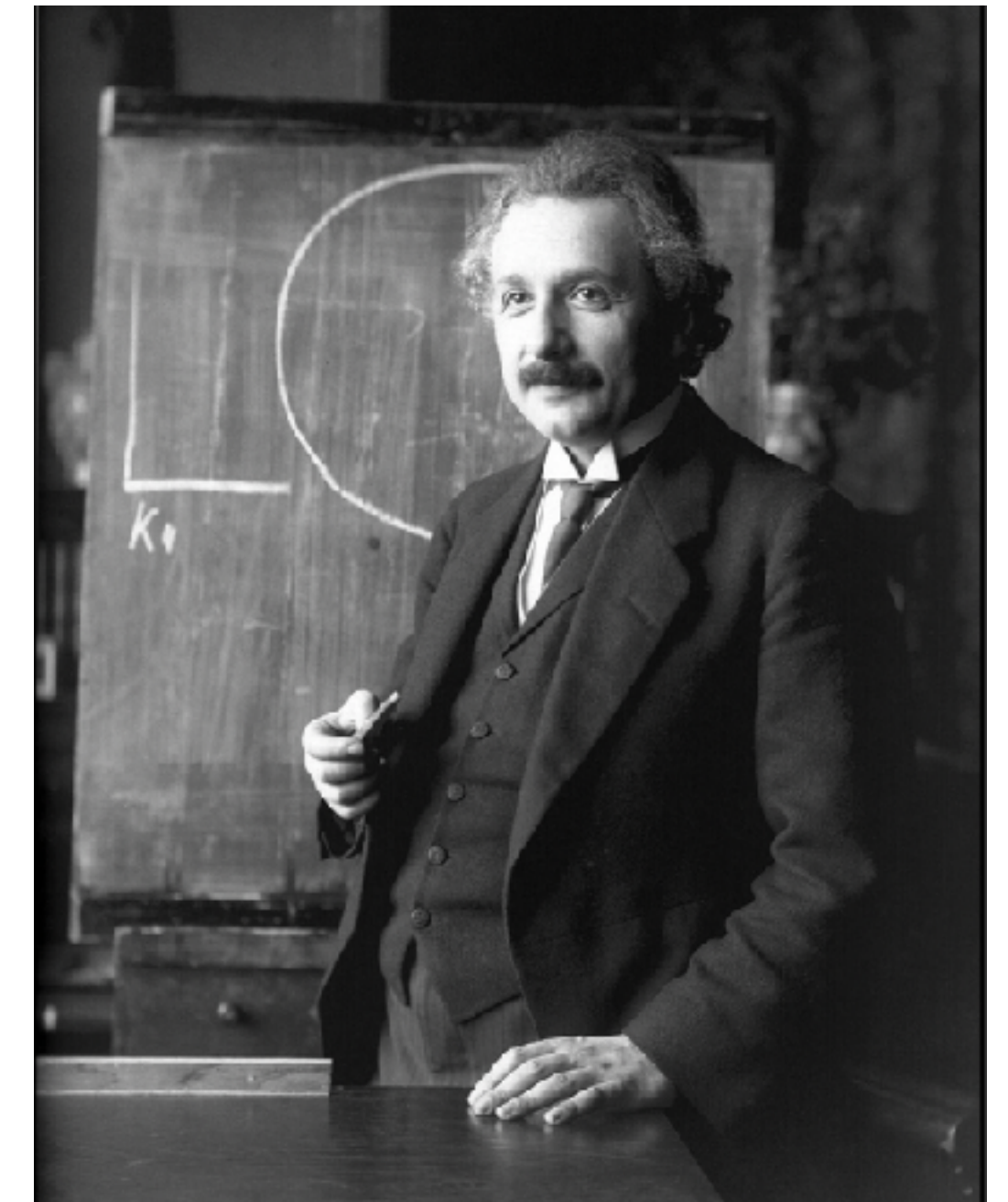
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Tidal forces (shear)



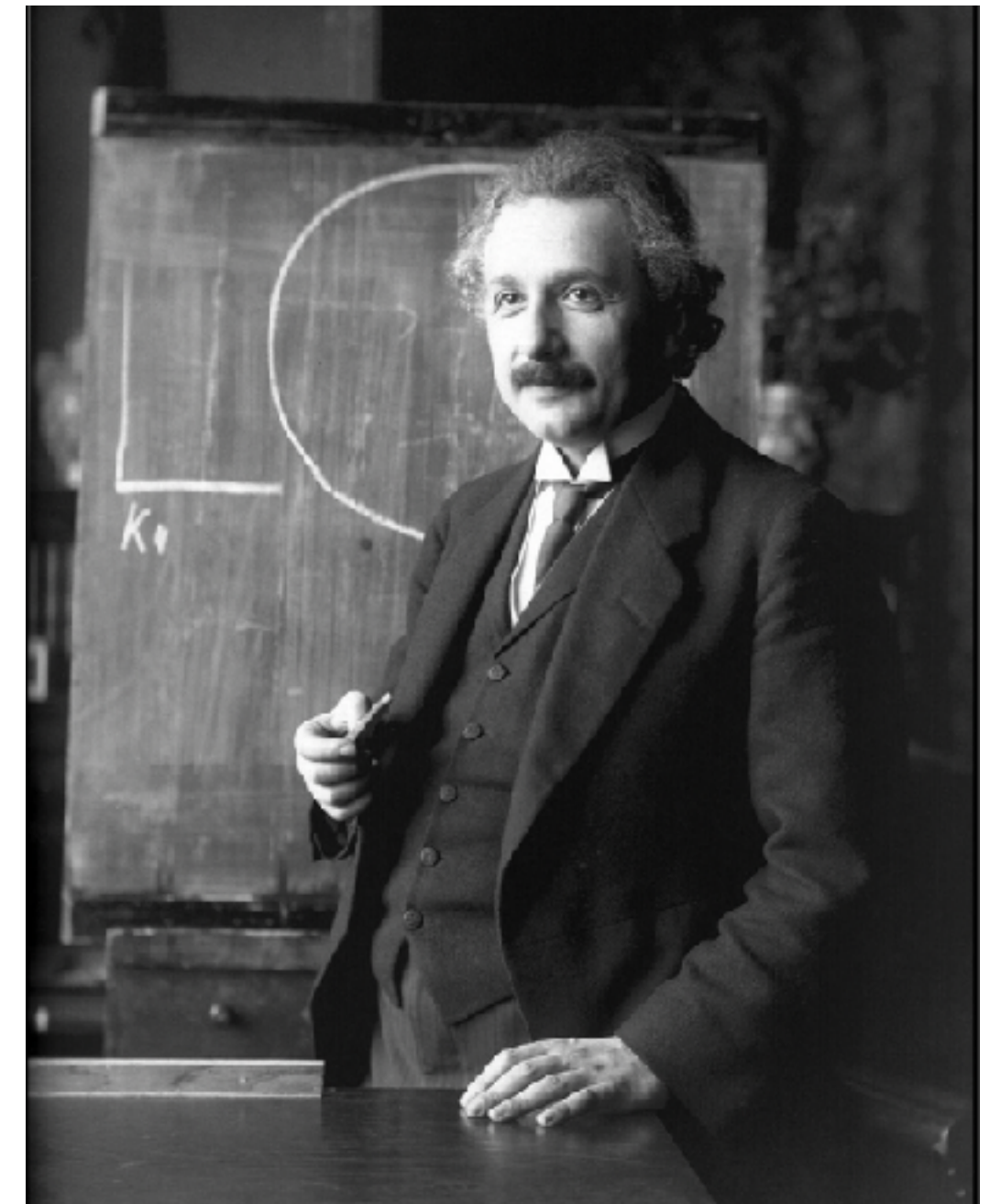
Albert Einstein (1879 - 1955)

TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

I have the metric tensor $g_{\mu\nu}$ which tells how to compute distances

I have the Energy/momentum tensor $T_{\mu\nu}$ which tells how matter/energy distribute in spacetime

I have the Riemann tensor $R^\lambda_{\rho\mu\nu}$ which tells how curved spacetime is



Albert Einstein (1879 - 1955)

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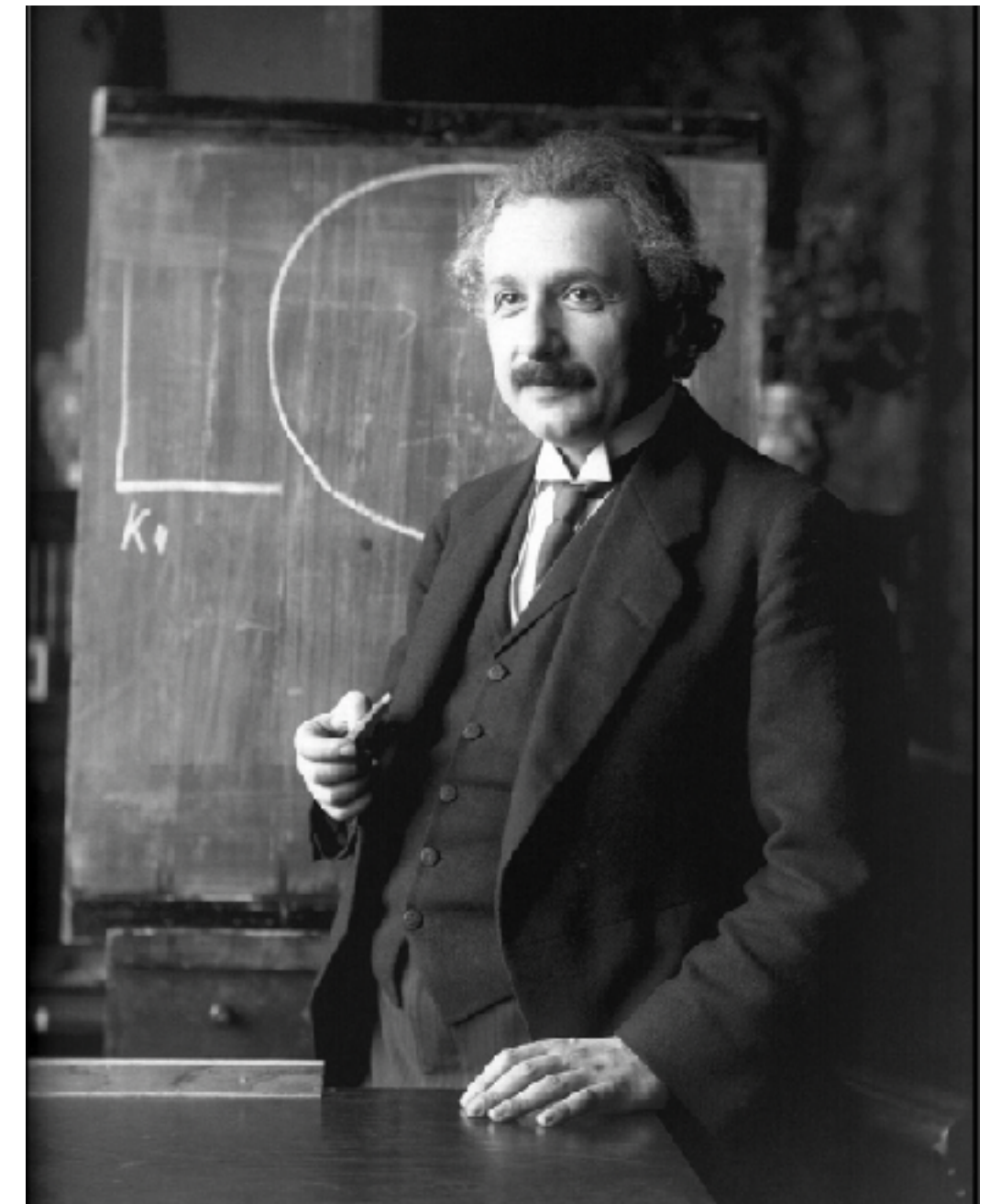
I have the Energy/momentum tensor $T_{\mu\nu}$ which tells how matter/energy distribute in spacetime

I have the Riemann tensor $R^\lambda_{\rho\mu\nu}$ which tells how curved spacetime is

Aha!

$$R^\lambda_{\rho\mu\nu} \propto T_{\mu\nu}$$

Obvious problem: different number of indices. Not quite there yet.

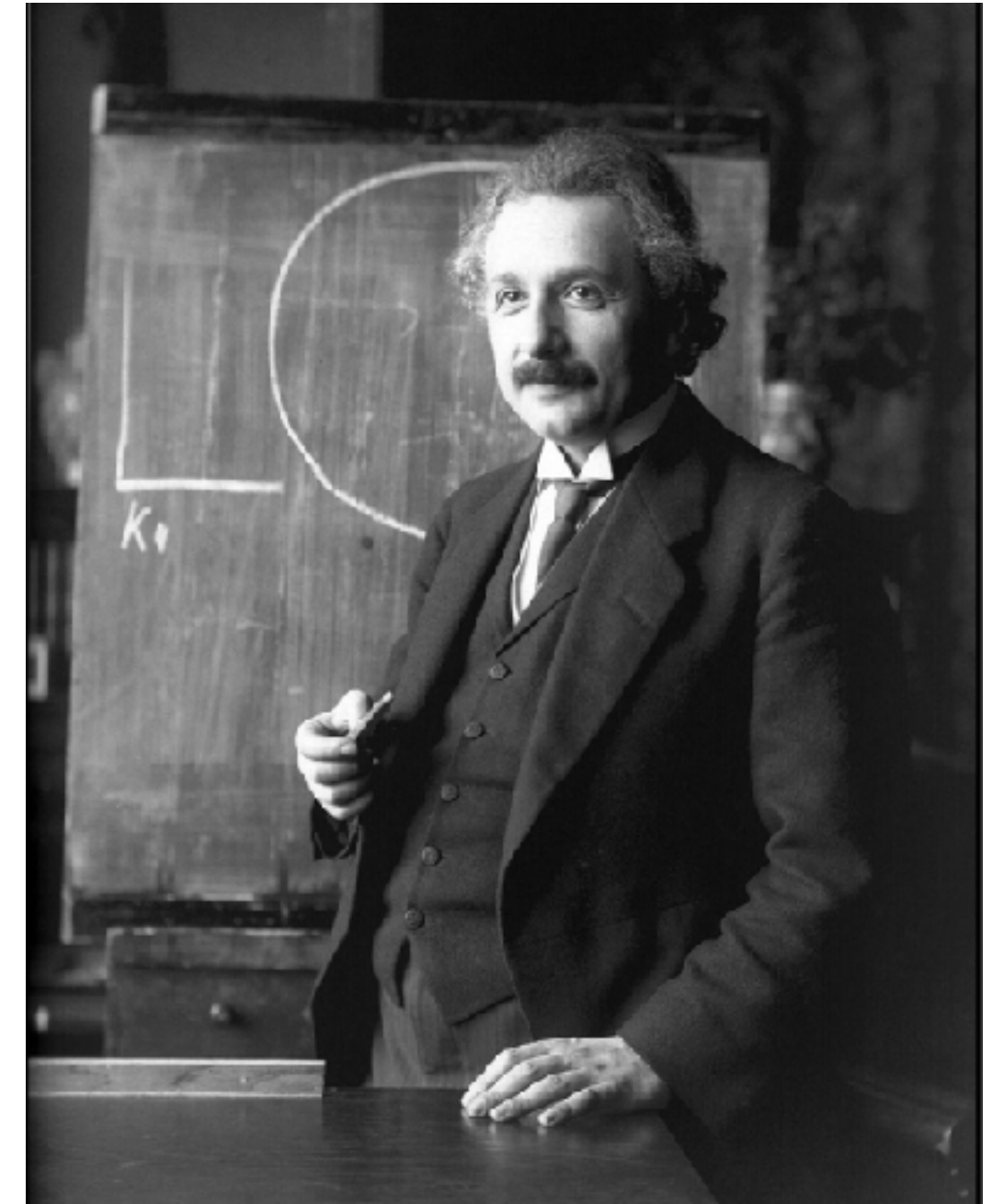


Albert Einstein (1879 - 1955)

TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

Some Italians found ways of **contracting** the Riemman tensor:

$$R^{\lambda}_{\rho\mu\nu} \rightarrow R_{\mu\nu} \rightarrow R$$



Albert Einstein (1879 - 1955)

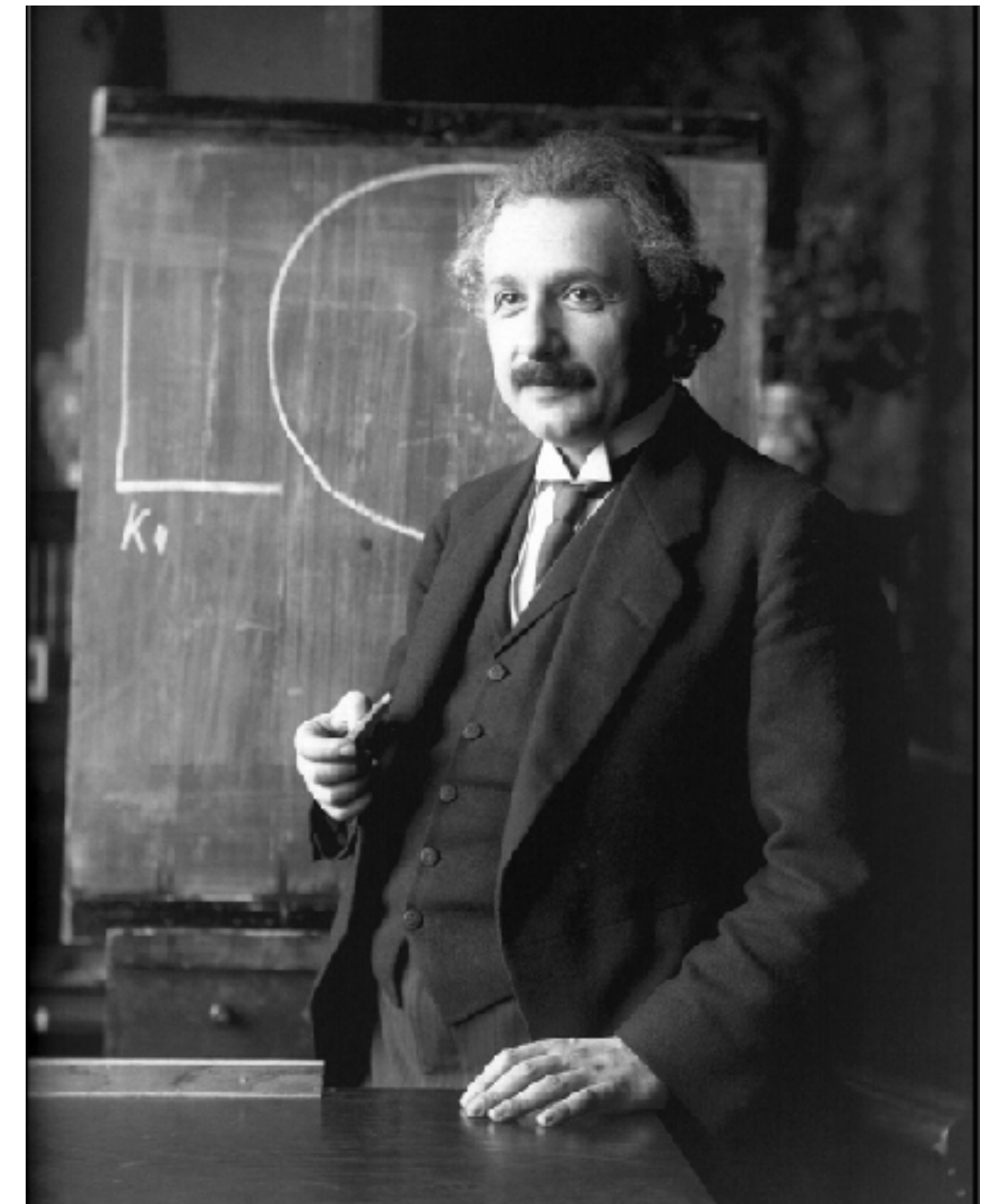
TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

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$$R^{\lambda}_{\rho\mu\nu} \rightarrow R_{\mu\nu} \rightarrow R$$

Now I get it!

$$R_{\mu\nu} \propto T_{\mu\nu}$$



Albert Einstein (1879 - 1955)

TOWARDS GENERAL RELATIVITY – EINSTEIN'S EQUATIONS (1915)

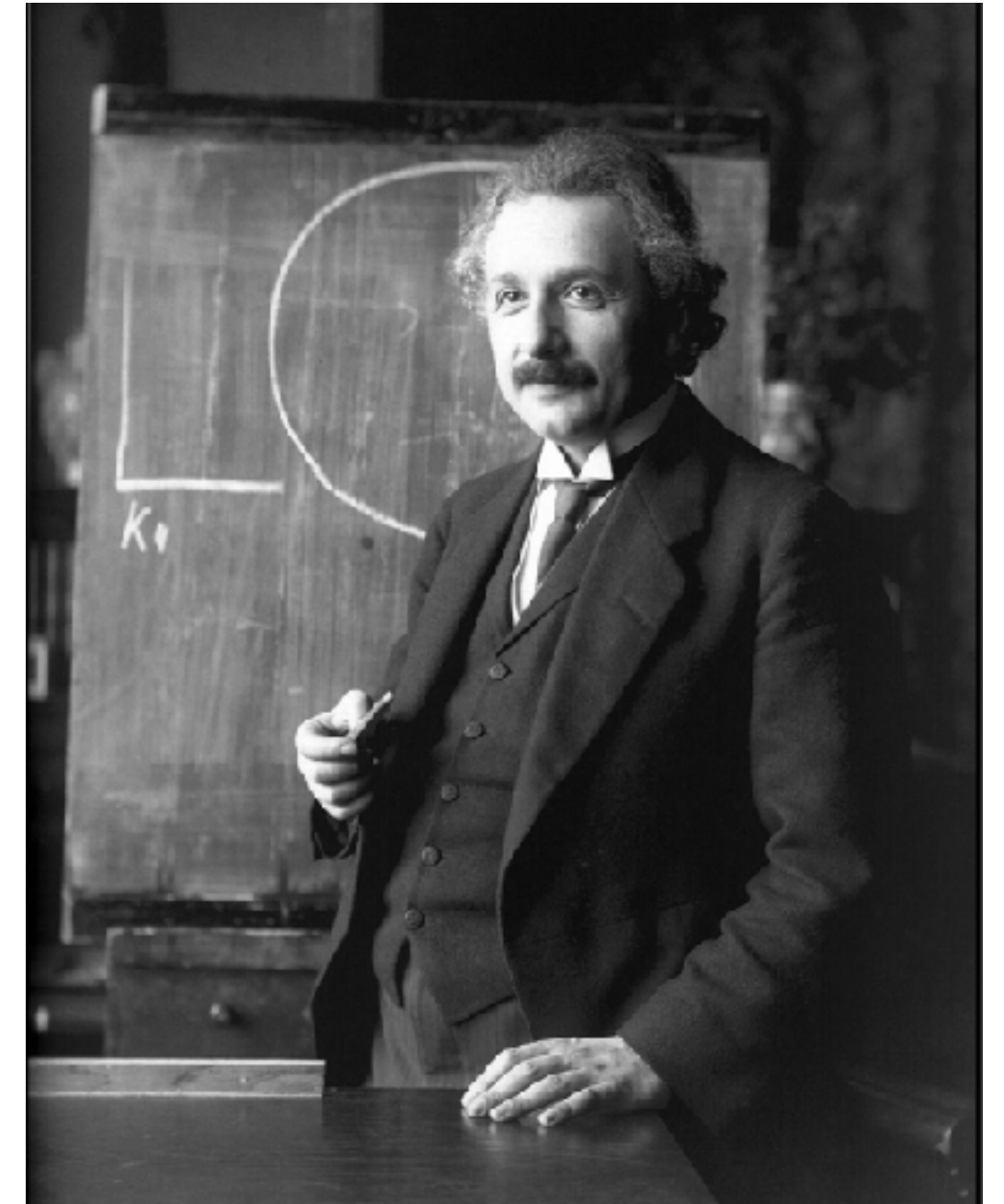
Some Italians found ways of **contracting** the Riemann tensor:

$$R^{\lambda}_{\rho\mu\nu} \rightarrow R_{\mu\nu} \rightarrow R$$

Now I get it!

~~$$R_{\mu\nu} \propto T_{\mu\nu}$$~~

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}$$



Albert Einstein (1879 - 1955)

Einstein's equations

TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

How to do cosmology using Einstein's equations?

$$\begin{aligned} R_{0102} = & g_{00} \left\{ \partial_0 \left[\frac{1}{2} (g^{00} (\partial_2 g_{10} + \partial_1 g_{20} - \partial_0 g_{12}) + g^{01} (\partial_2 g_{11} + \partial_1 g_{21} - \partial_1 g_{12}) + g^{02} (\partial_2 g_{12} + \partial_1 g_{22} - \partial_2 g_{12}) + g^{03} (\partial_2 g_{13} + \partial_1 g_{23} - \partial_3 g_{12})) \right] \right. \\ & - \partial_2 \left[\frac{1}{2} (g^{00} (\partial_0 g_{10} + \partial_1 g_{00} - \partial_0 g_{01}) + g^{01} (\partial_0 g_{11} + \partial_1 g_{01} - \partial_1 g_{01}) + g^{02} (\partial_0 g_{12} + \partial_1 g_{02} - \partial_2 g_{01}) + g^{03} (\partial_0 g_{13} + \partial_1 g_{03} - \partial_3 g_{01})) \right] \\ & + \left[\frac{1}{2} (g^{00} (\partial_0 g_{00} + \partial_0 g_{00} - \partial_0 g_{00}) + g^{01} (\partial_0 g_{01} + \partial_1 g_{00} - \partial_1 g_{00}) + g^{02} (\partial_0 g_{02} + \partial_2 g_{00} - \partial_2 g_{00}) + g^{03} (\partial_0 g_{03} + \partial_3 g_{00} - \partial_3 g_{00})) \right] \\ & \times \left[\frac{1}{2} (g^{00} (\partial_2 g_{10} + \partial_1 g_{20} - \partial_0 g_{12}) + g^{01} (\partial_2 g_{11} + \partial_1 g_{21} - \partial_1 g_{12}) + g^{02} (\partial_2 g_{12} + \partial_1 g_{22} - \partial_2 g_{12}) + g^{03} (\partial_2 g_{13} + \partial_1 g_{23} - \partial_3 g_{12})) \right] \\ & - \left[\frac{1}{2} (g^{00} (\partial_2 g_{00} + \partial_0 g_{20} - \partial_0 g_{20}) + g^{01} (\partial_2 g_{01} + \partial_1 g_{20} - \partial_1 g_{20}) + g^{02} (\partial_2 g_{02} + \partial_2 g_{20} - \partial_2 g_{20}) + g^{03} (\partial_2 g_{03} + \partial_3 g_{20} - \partial_3 g_{20})) \right] \\ & \times \left[\frac{1}{2} (g^{00} (\partial_0 g_{10} + \partial_1 g_{00} - \partial_0 g_{01}) + g^{01} (\partial_0 g_{11} + \partial_1 g_{01} - \partial_1 g_{01}) + g^{02} (\partial_0 g_{12} + \partial_1 g_{02} - \partial_2 g_{01}) + g^{03} (\partial_0 g_{13} + \partial_1 g_{03} - \partial_3 g_{01})) \right] \Big\} \\ & + g_{01} \left\{ \text{(same structure with } g^{0*} \rightarrow g^{1*}) \right\} + g_{02} \left\{ \text{(same structure with } g^{0*} \rightarrow g^{2*}) \right\} + g_{03} \left\{ \text{(same structure with } g^{0*} \rightarrow g^{3*}) \right\} \end{aligned}$$

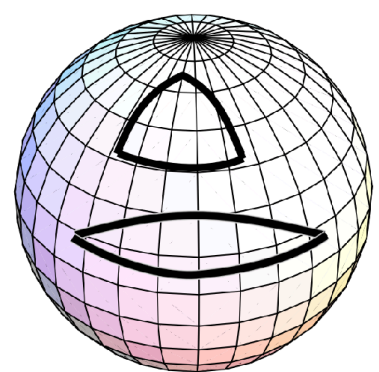
TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

How to do cosmology using Einstein's equations?

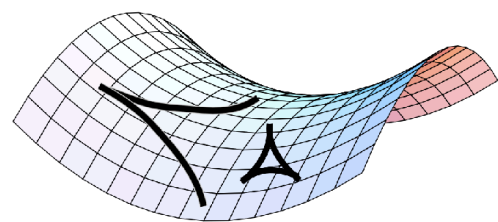
The Universe is *spatially* homogeneous and evolve in time:

$$ds^2 = -dt^2 + a(t)^2 \left[\frac{dr^2}{1 - \kappa r^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

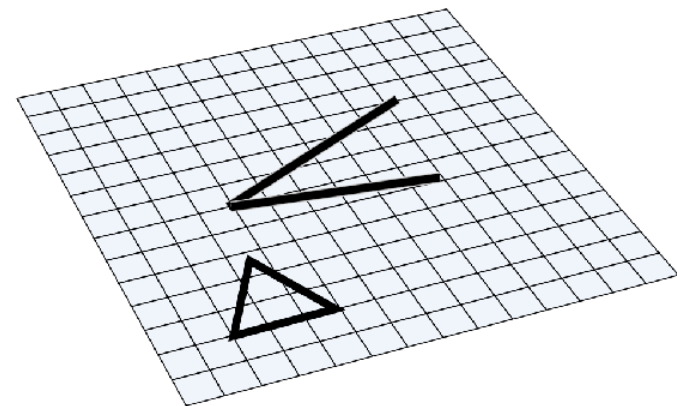
Friedmann–Lemaître–Robertson–Walker metric (FLRW)



$\kappa > 0$

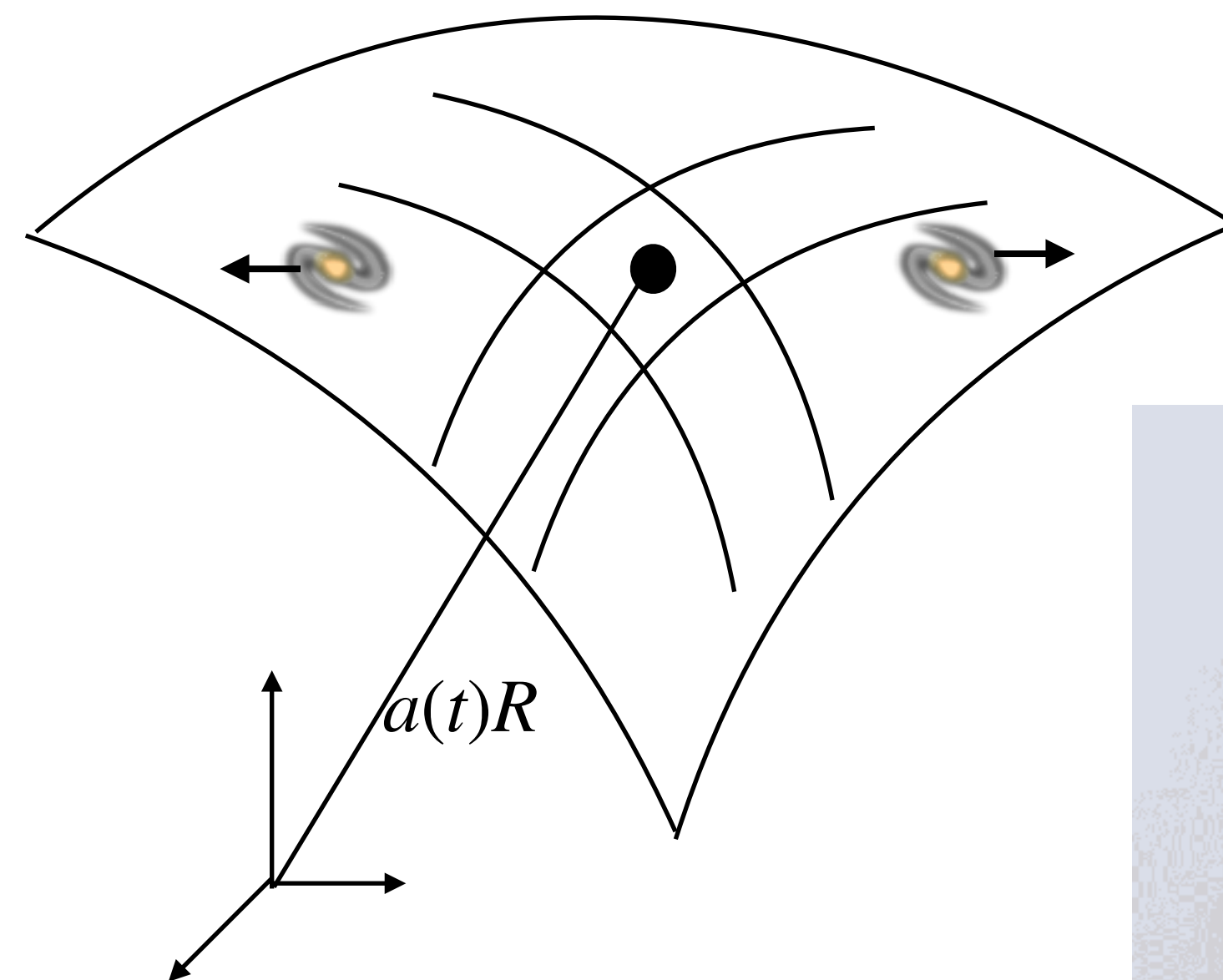


$\kappa = 0$



$\kappa < 0$

$a(t)$ Scale factor: How distances are changing with time



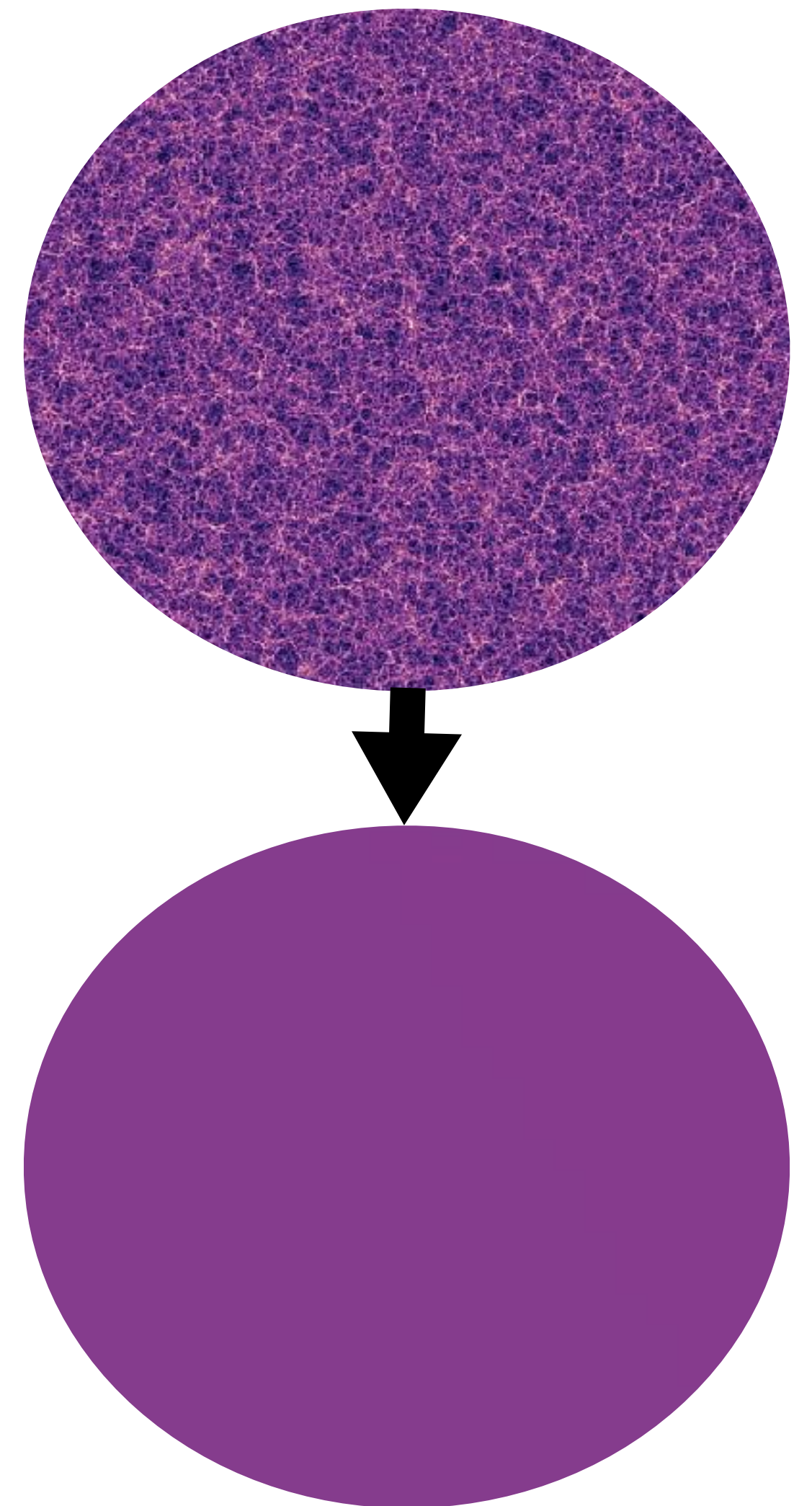
TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

Energy-momentum tensor of an homogeneous and isotropic fluid:

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu - pg^{\mu\nu} \quad u^\mu = (-1, 0, 0, 0)$$

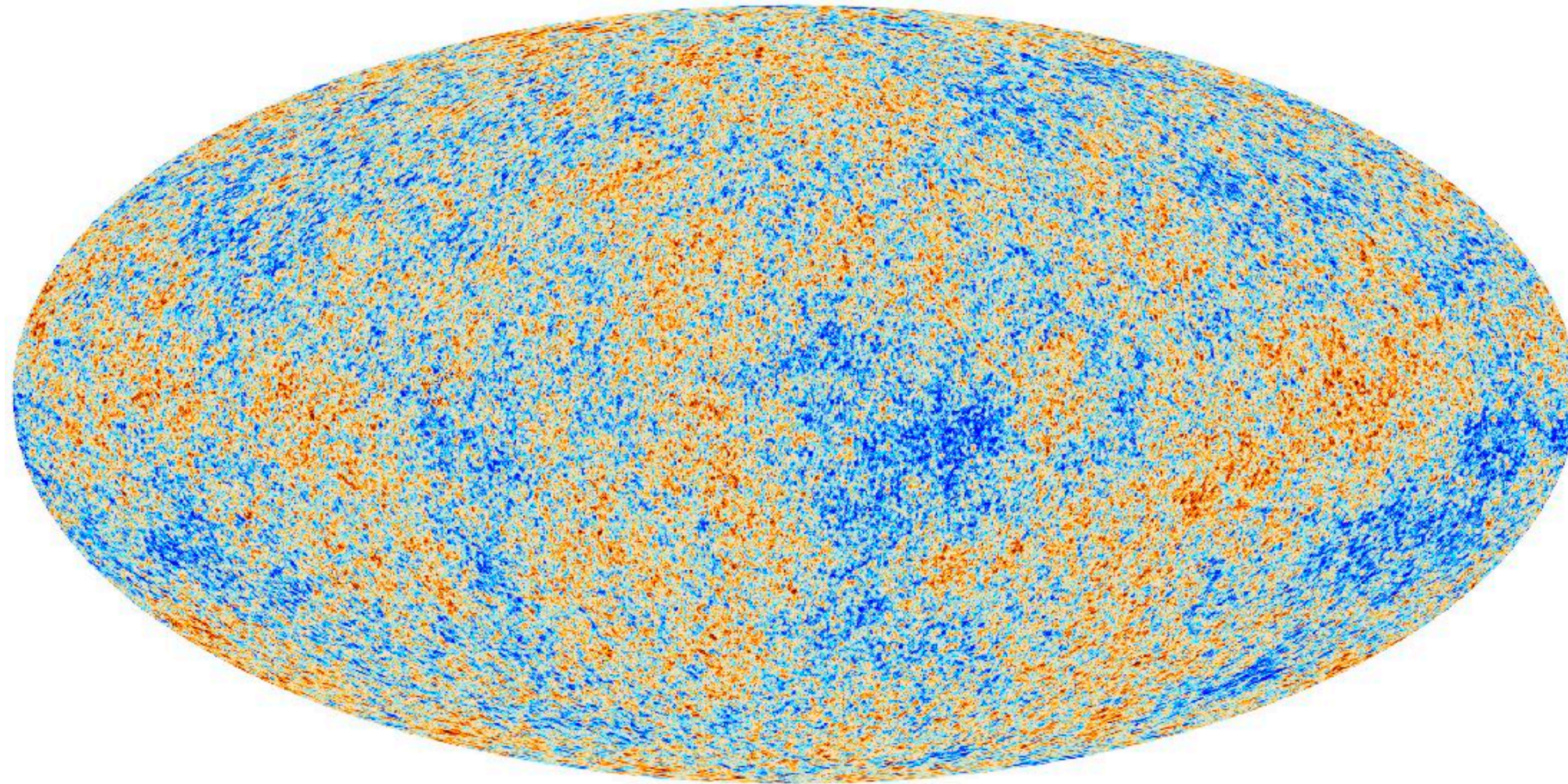
$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & -p & 0 & 0 \\ 0 & 0 & -p & 0 \\ 0 & 0 & 0 & -p \end{pmatrix}$$

$\rho(t), p(t)$ Homogenous density/pressure



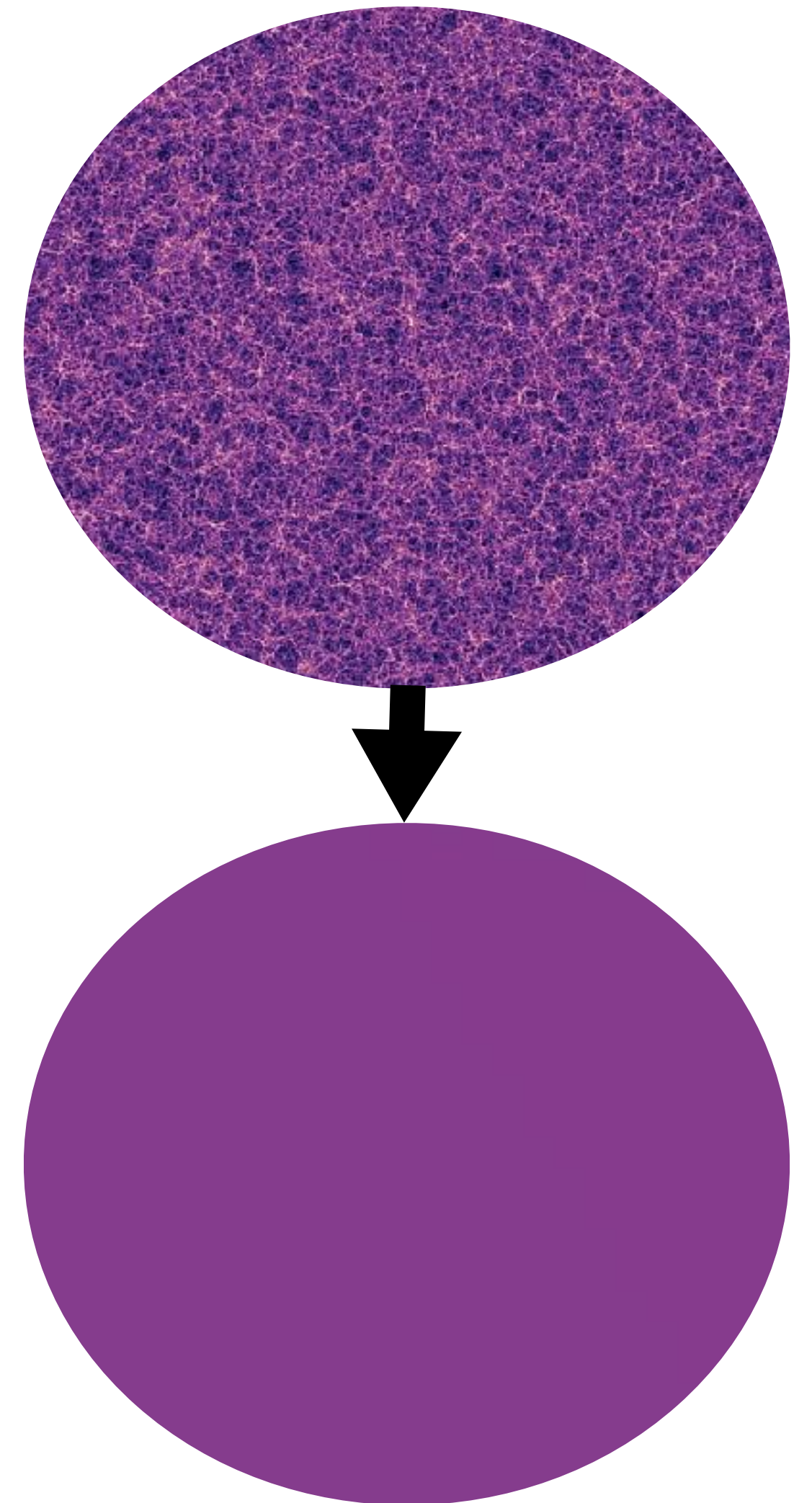
TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

Actually, this idea is not that far from reality:



Credit: Planck collaboration

The maximum deviation from the average is $\simeq 10^{-5}$!!

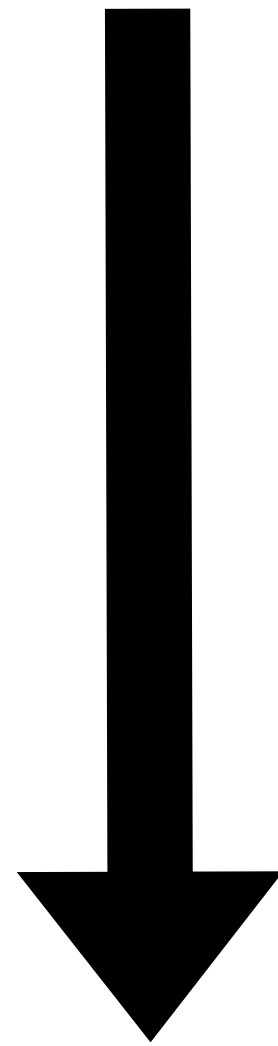


TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}$$

Homogeneous and isotropic

$g_{\mu\nu}$ and $T_{\mu\nu}$

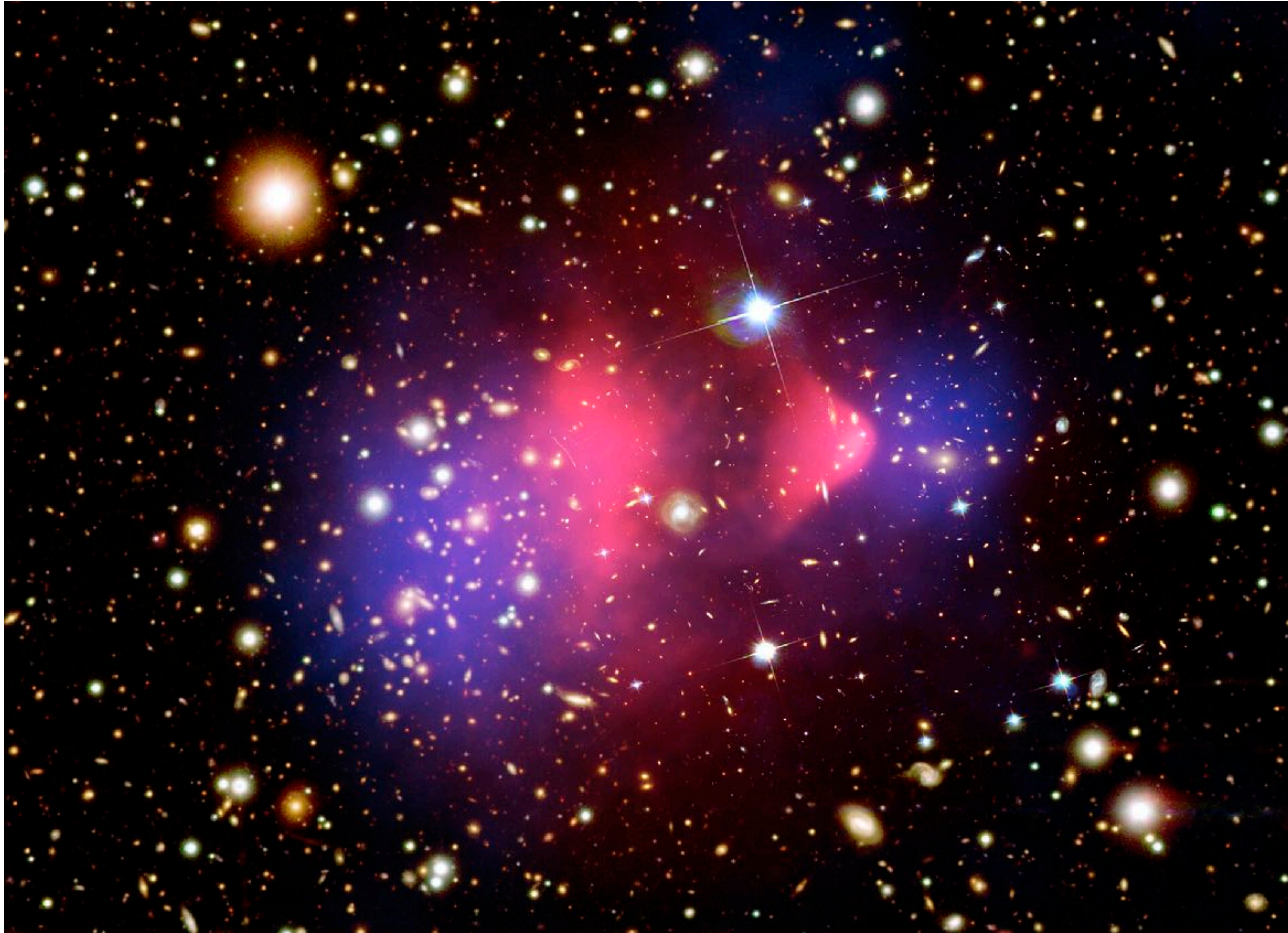


$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p)$$

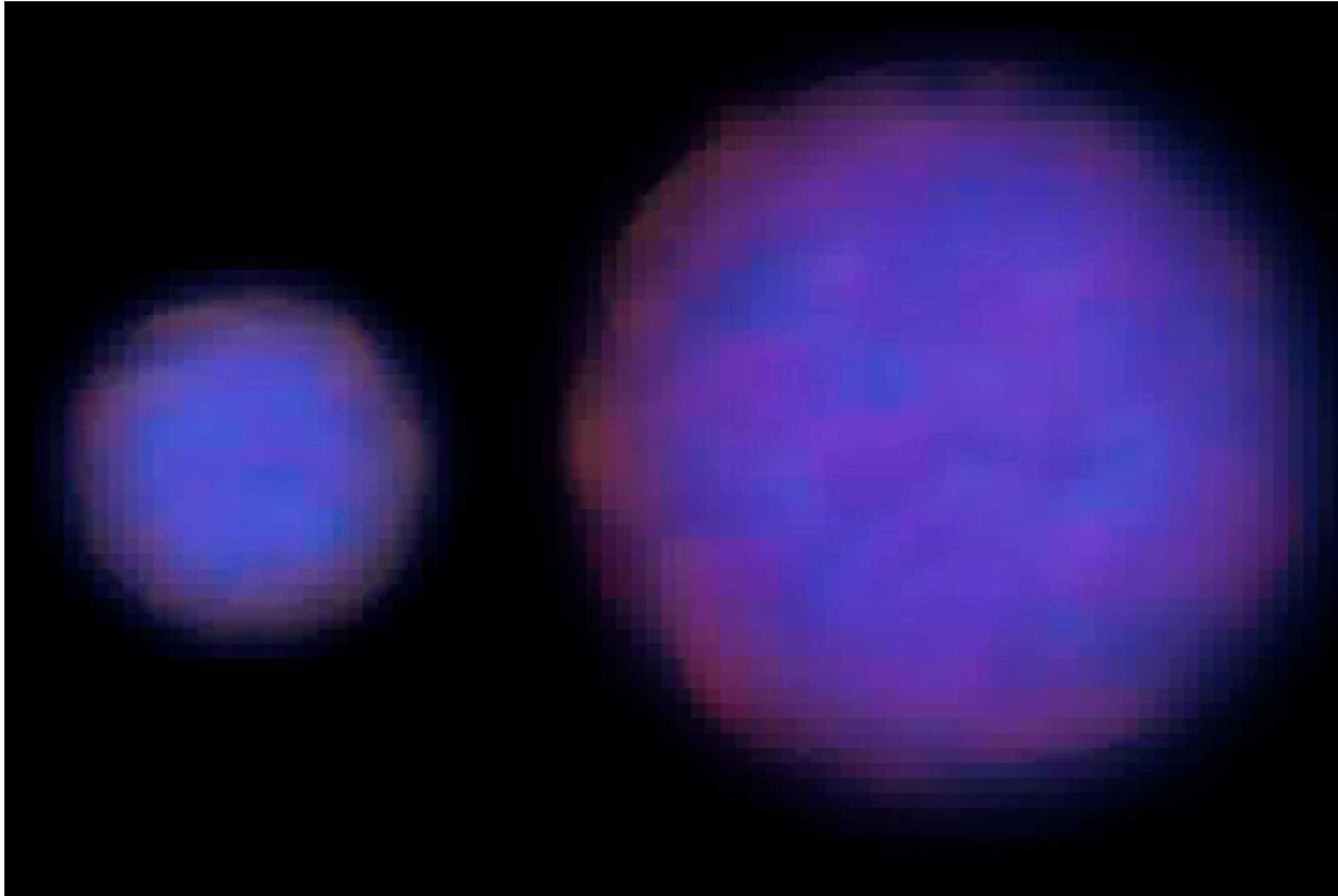
Friedmann equations

WHAT IS THE UNIVERSE MADE OF – DARK MATTER (THE BULLET CLUSTER)



Credit: ESA

WHAT IS THE UNIVERSE MADE OF – DARK MATTER (THE BULLET CLUSTER)



Credit: <https://www.youtube.com/watch?v=eC5Lwjsgl4I>

WHAT IS THE UNIVERSE MADE OF – DARK MATTER

$t = 0.2 \text{ Gyr}$

A large, solid blue rectangle occupies the central portion of the slide. It is a uniform, deep blue color with no visible features, patterns, or text inside it. This likely represents a visualization of a dark matter distribution or a simulation snapshot at the time specified by the text.

WHAT IS THE UNIVERSE MADE OF – DARK ENERGY

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \boxed{\Lambda g_{\mu\nu}} = 8\pi GT_{\mu\nu}$$

Einstein's equations allow for a constant term

The Universe (and Switzerland) is filled with fields



TOWARDS COSMOLOGY – FRIEDMANN EQUATIONS

$$\left(\frac{\dot{a}}{a}\right)^2 = \sum_i \frac{8\pi G}{3} \rho_i$$

$$\frac{\ddot{a}}{a} = \sum_i -\frac{4\pi G}{3} (\rho_i + 3p_i)$$

Cold Dark Matter:

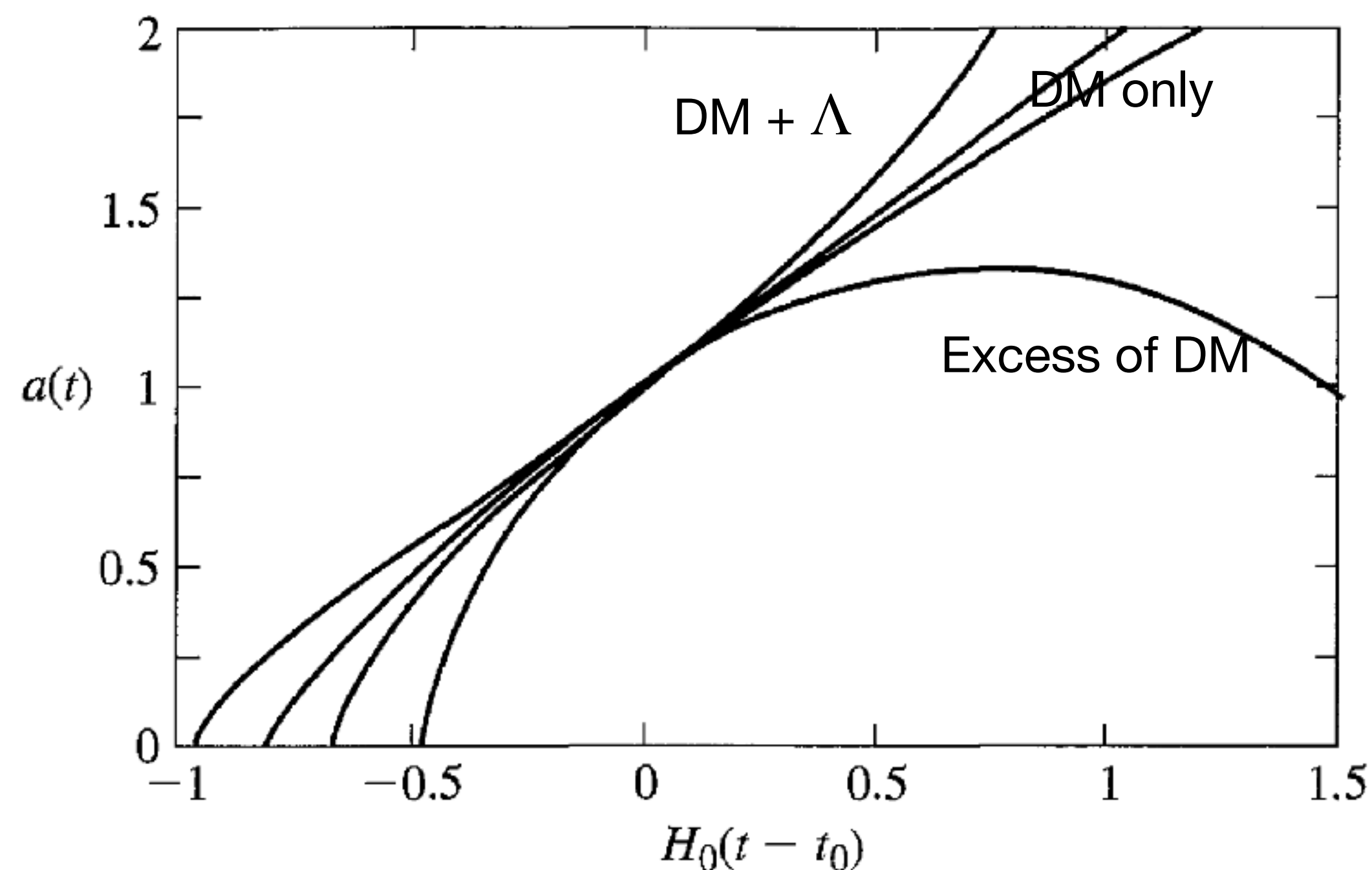
$$\rho_m \propto a^{-3} \quad p_m = 0$$

Dark Energy:

$$\rho_\Lambda \propto \exp(H_0 t) \quad p_\Lambda = -\rho_\Lambda$$

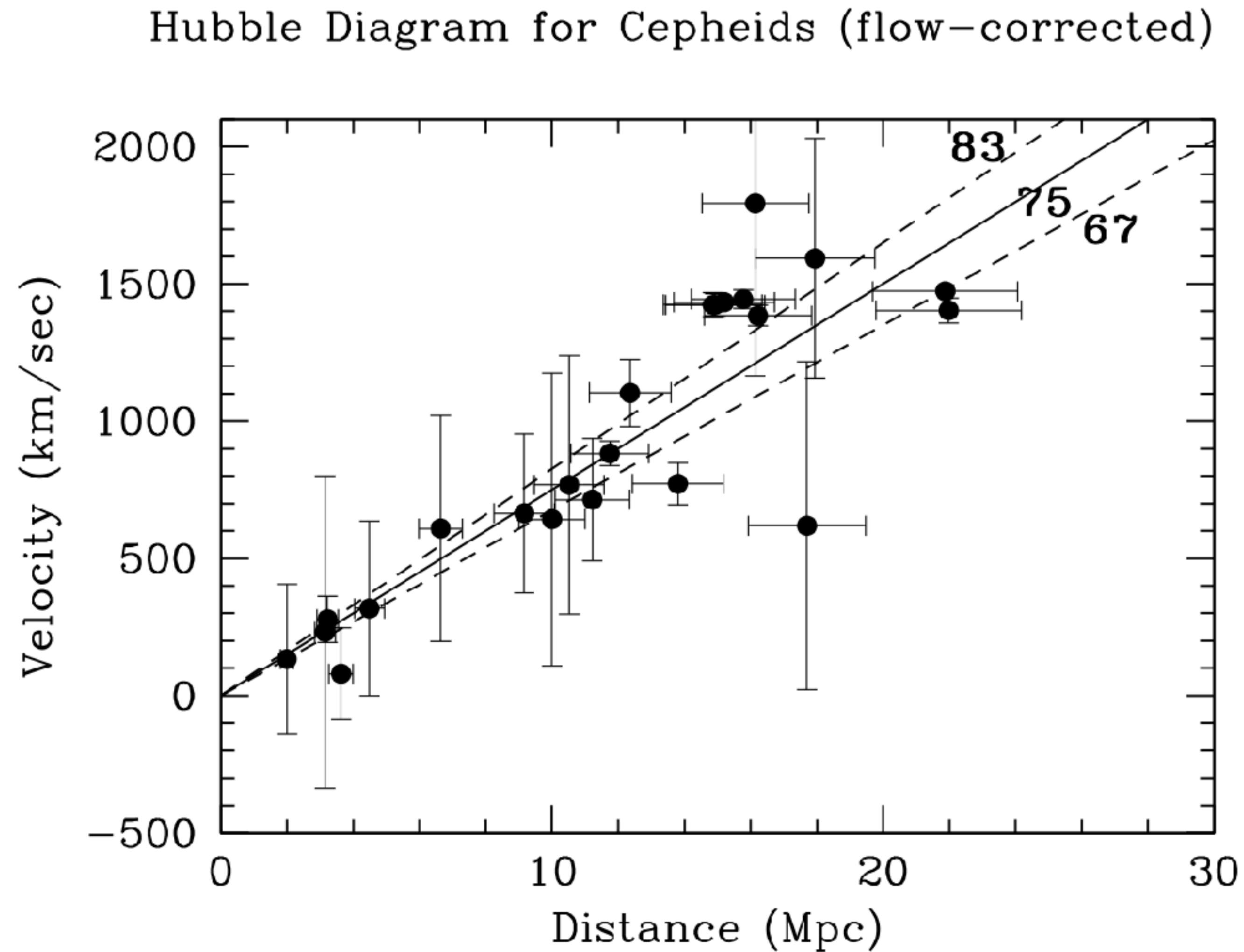
Radiation:

$$\rho_r \propto a^{-4} \quad p_r = -\frac{1}{3}\rho_r$$



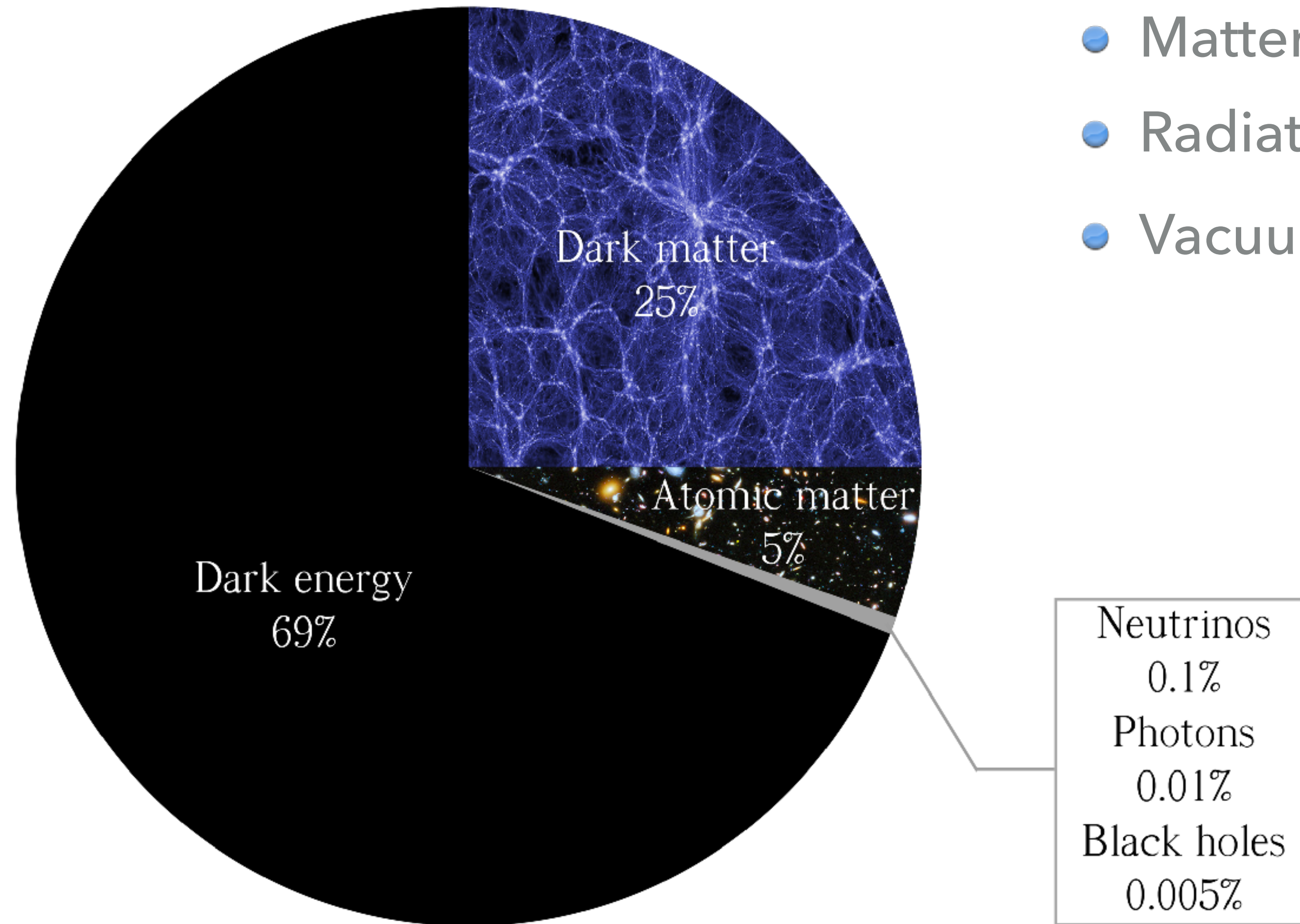
The Einstein's equations predicts an expanding Universe!

THE EXPANDING UNIVERSE – HUBBLE'S LAW (1929)



$$v = H_0 d \quad H_0 \simeq 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

WHAT IS THE UNIVERSE MADE OF – MATTER/ENERGY CONTENT



- Matter: moves slowly ($v \ll c=1$)
- Radiation: $v=c=1$ (massless)
- Vacuum (Dark Energy)